Charting New Paths in Lipidomics: Understanding Caloric Restriction and Aging with Orbitrap Astral MS

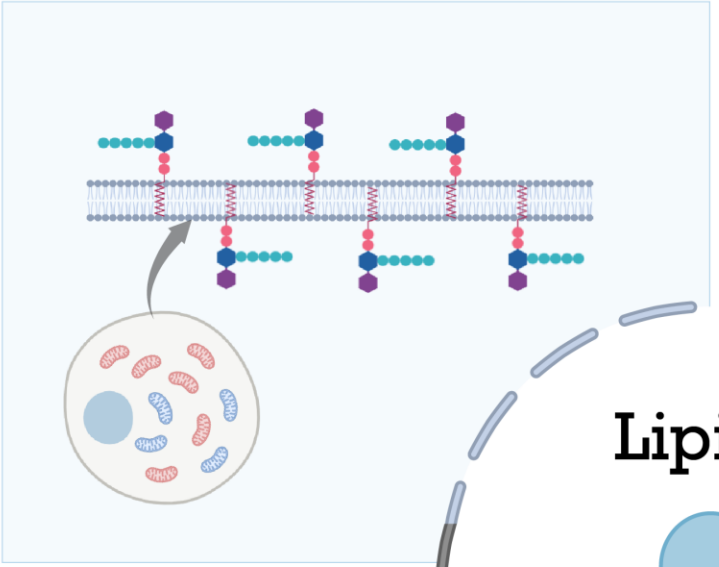
Salma Abou Elhassan, Katherine Overmyer, Benton Anderson, Scott Quarmby, Josef Clark, Rozalyn Anderson, Ricki Colman, and Joshua Coon

University of Wisconsin-Madison; Morgridge Institute for Research; National Center for Quantitative Biology of Complex Systems; and Wisconsin National Primate Research Center, Madison, WI

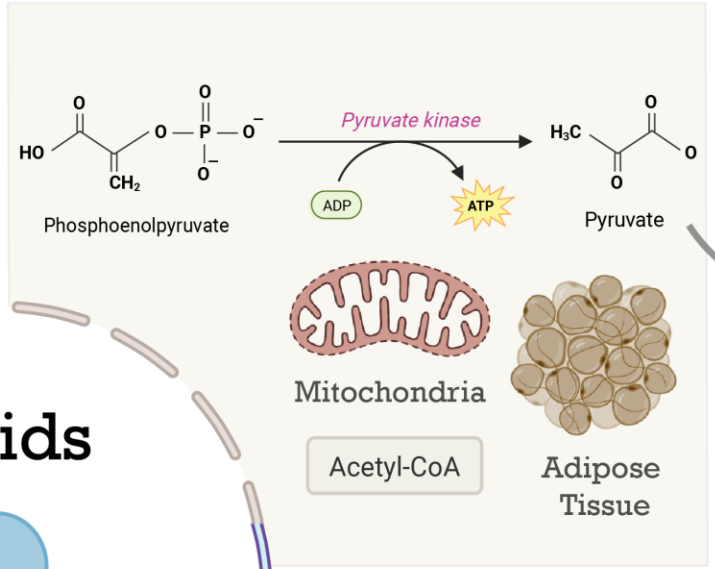
Conflict of Interest: The authors declare the following financial interests: Joshua Coon is a consultant for Thermo Fisher Scientific, Seer, and a co-founder of CelermAb Inc

Lipids Play Key Roles in Many Biological Processes

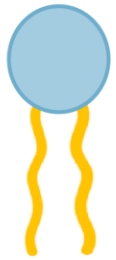
Cell Membrane Structure



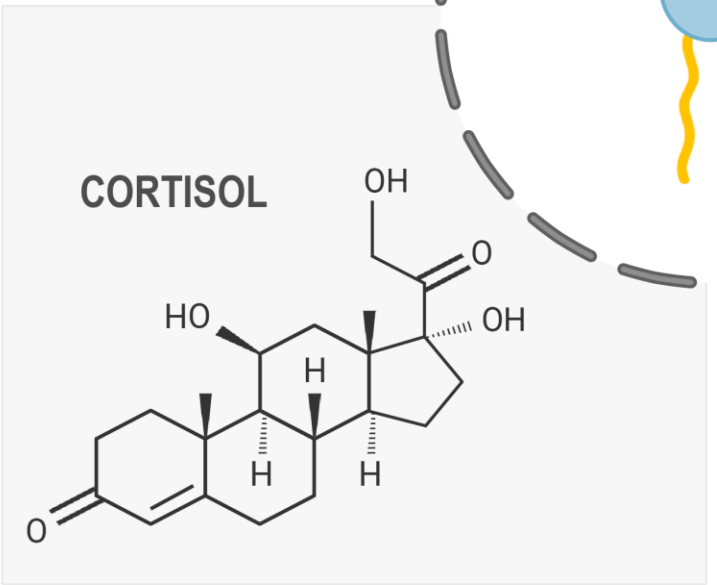
Energy Storage and Production



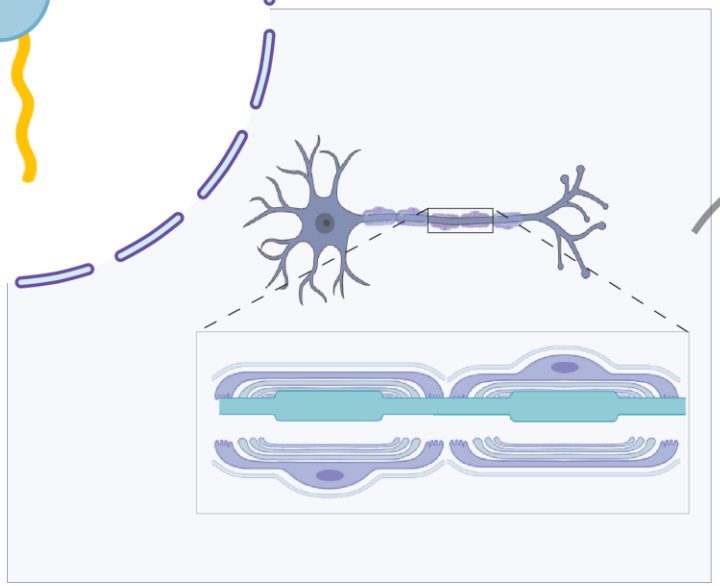
Lipids



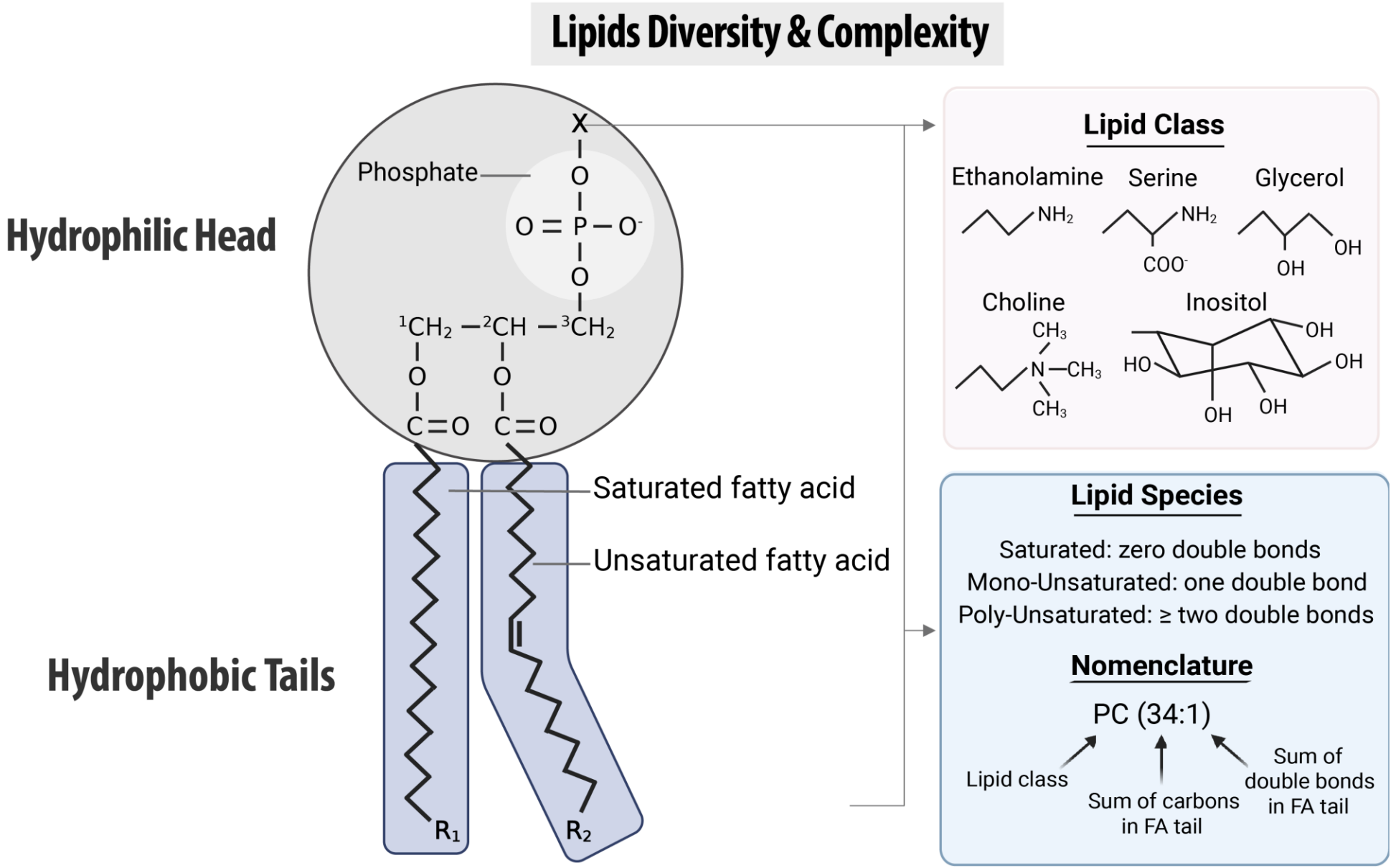
Hormonal Regulation



Neural Cell Protection &
Nerve Impulse Transmission



Lipids are Highly Diverse in Structure

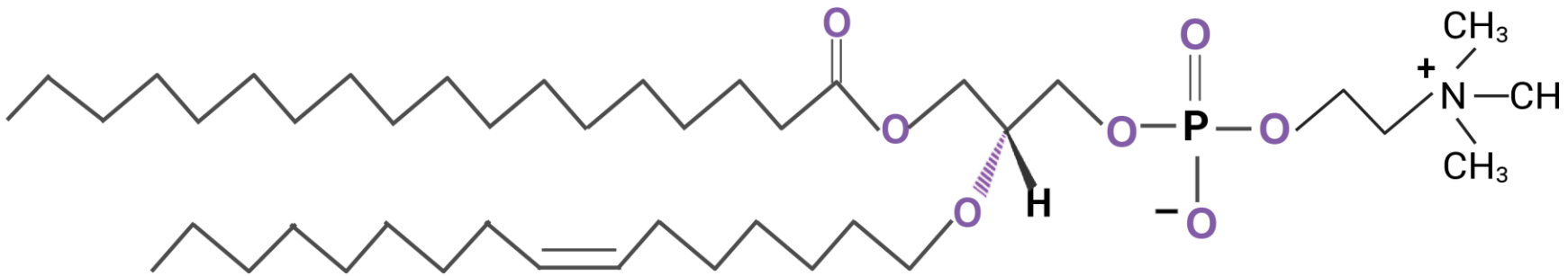


Isomers and Isobars are Key Sources of Complexity in Lipidomics

Isomers and Isobars

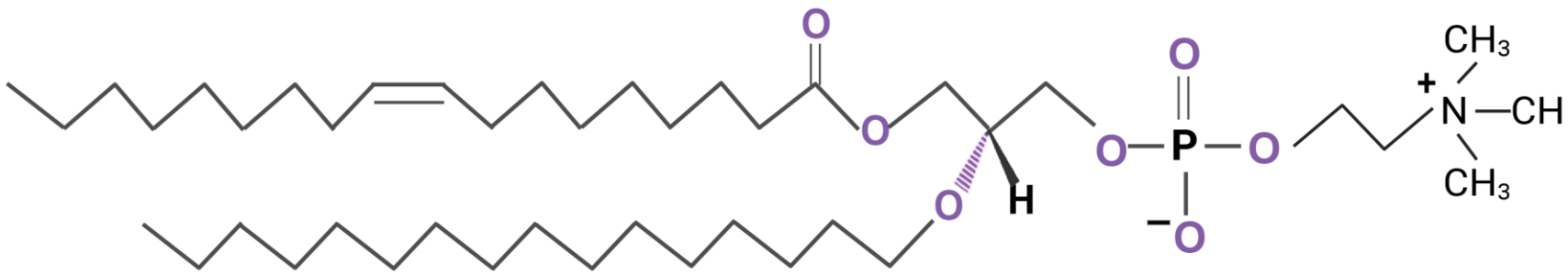
Having the exact same elemental composition, but different chemical structures

PC 16:1/18:0

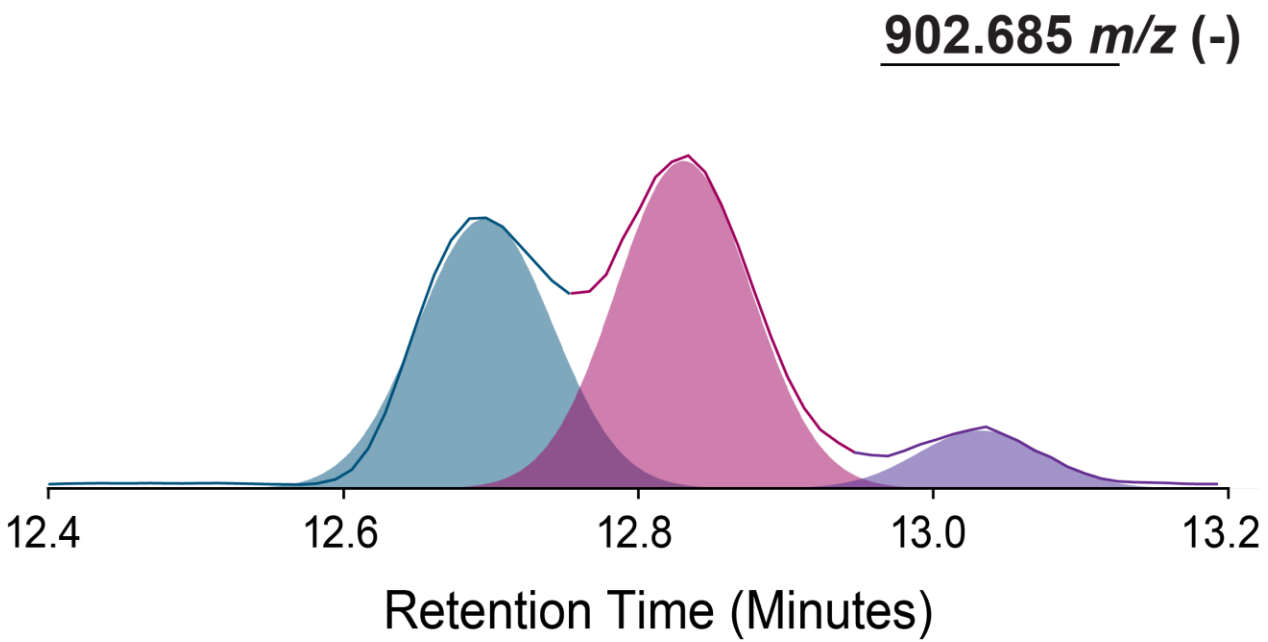


[M+H]+
760.5851 m/z

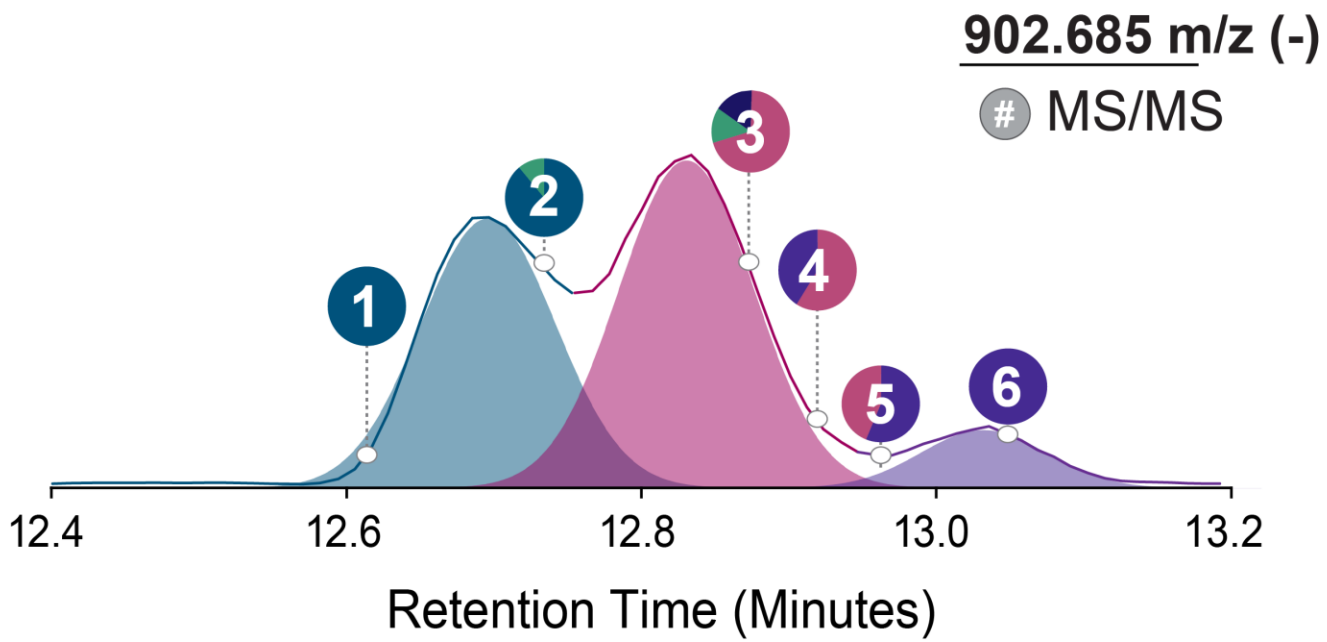
PC 16:0/18:1



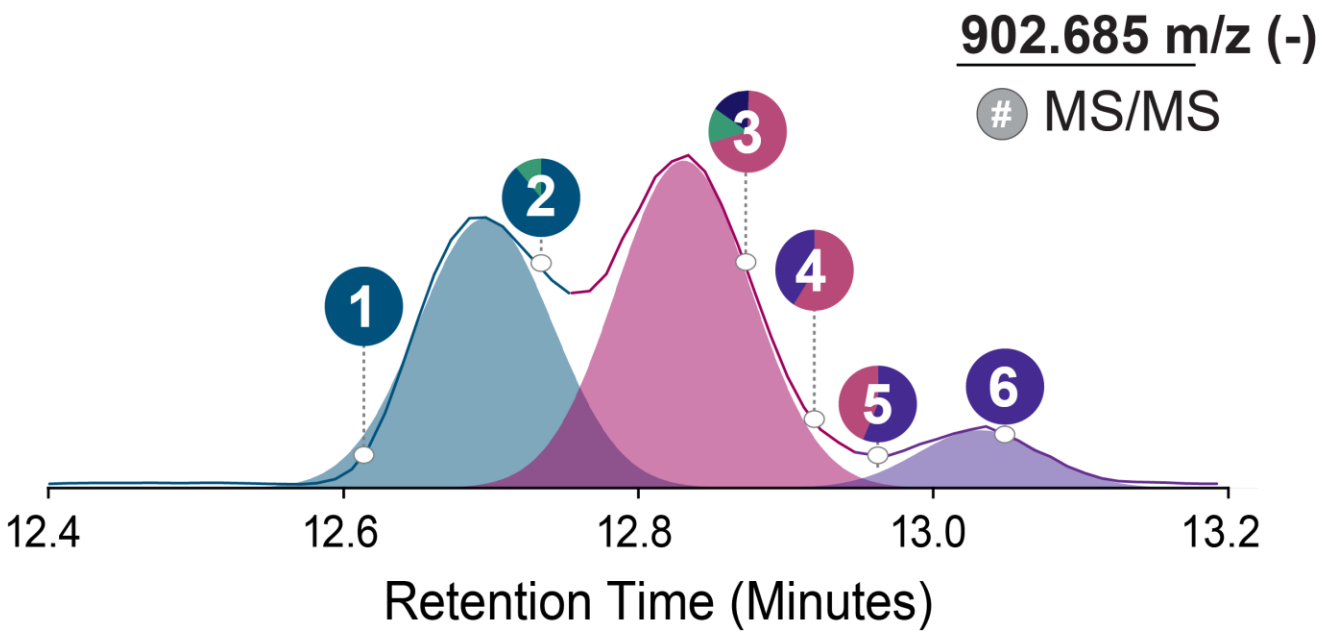
Isomers and Isobars are Key Sources of Complexity in Lipidomics



Isomers and Isobars are Key Sources of Complexity in Lipidomics



Isomers and Isobars are Key Sources of Complexity in Lipidomics

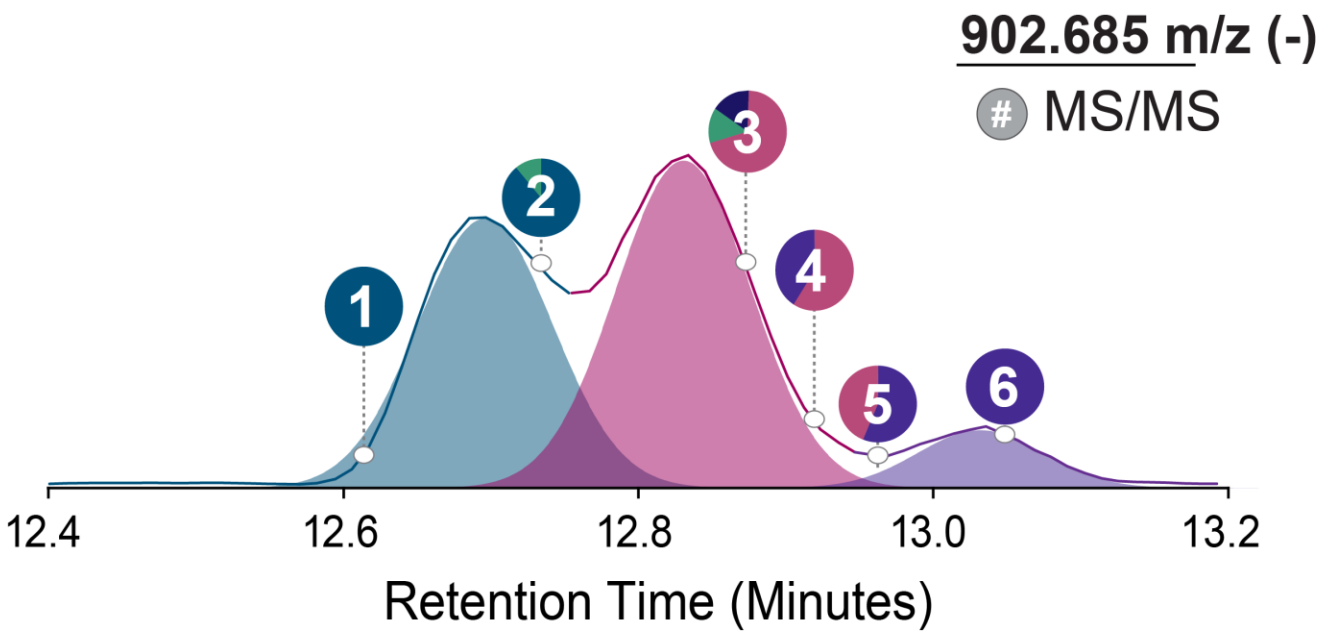


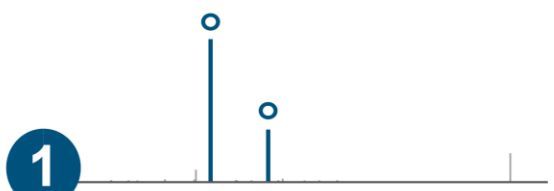


Spectral Purity

Lipid	Percent
PC 16:0_24:1	100%
PC 18:0_22:1	0%
PC 18:1_22:0	0%
PC 16:1_24:0	0%

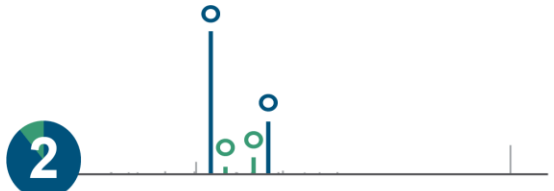
Isomers and Isobars are Key Sources of Complexity in Lipidomics





Spectral Purity

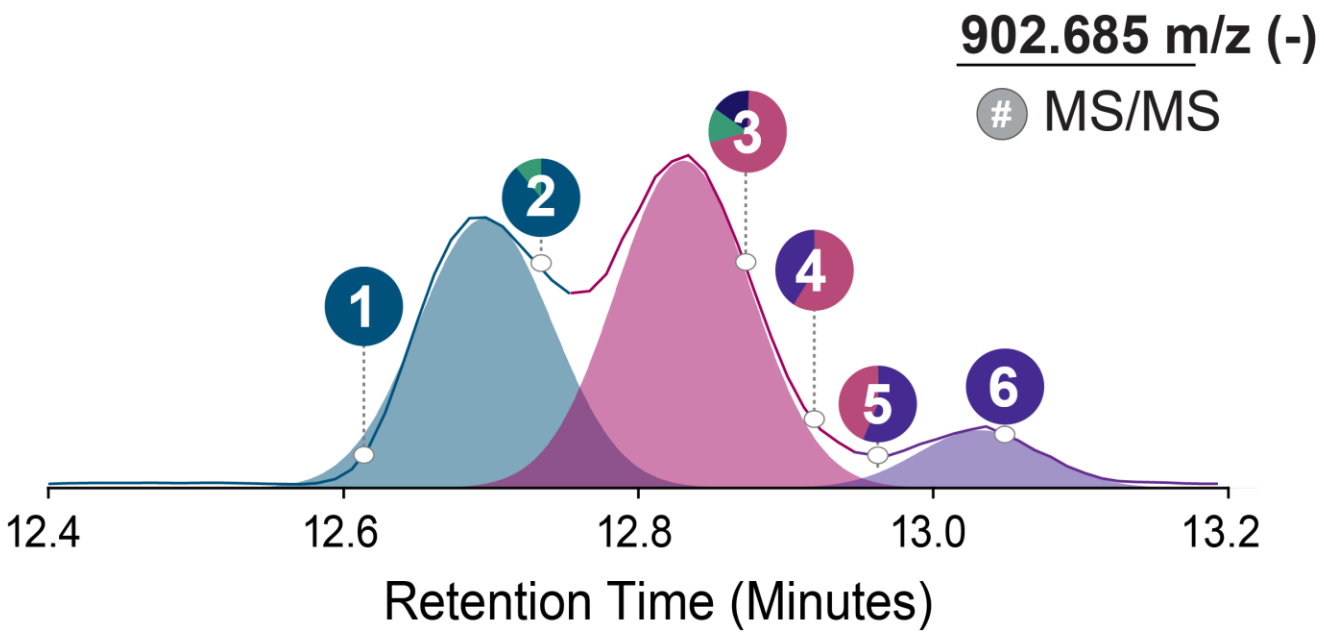
Lipid	Percent
PC 16:0_24:1	100%
PC 18:0_22:1	0%
PC 18:1_22:0	0%
PC 16:1_24:0	0%

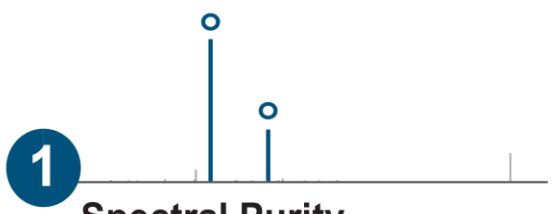


Spectral Purity

Lipid	Percent
PC 16:0_24:1	89%
PC 18:0_22:1	11%
PC 18:1_22:0	0%
PC 16:1_24:0	0%

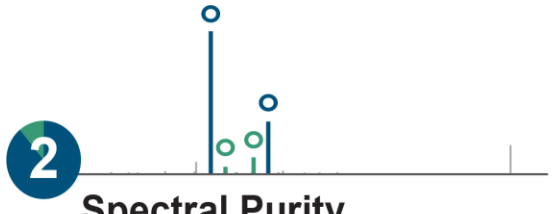
Isomers and Isobars are Key Sources of Complexity in Lipidomics





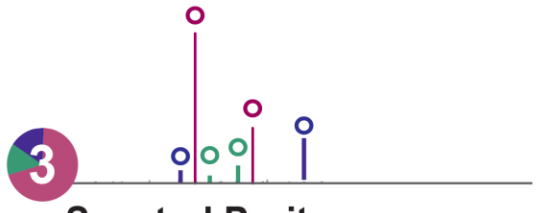
Spectral Purity

Lipid	Percent
PC 16:0_24:1	100%
PC 18:0_22:1	0%
PC 18:1_22:0	0%
PC 16:1_24:0	0%



Spectral Purity

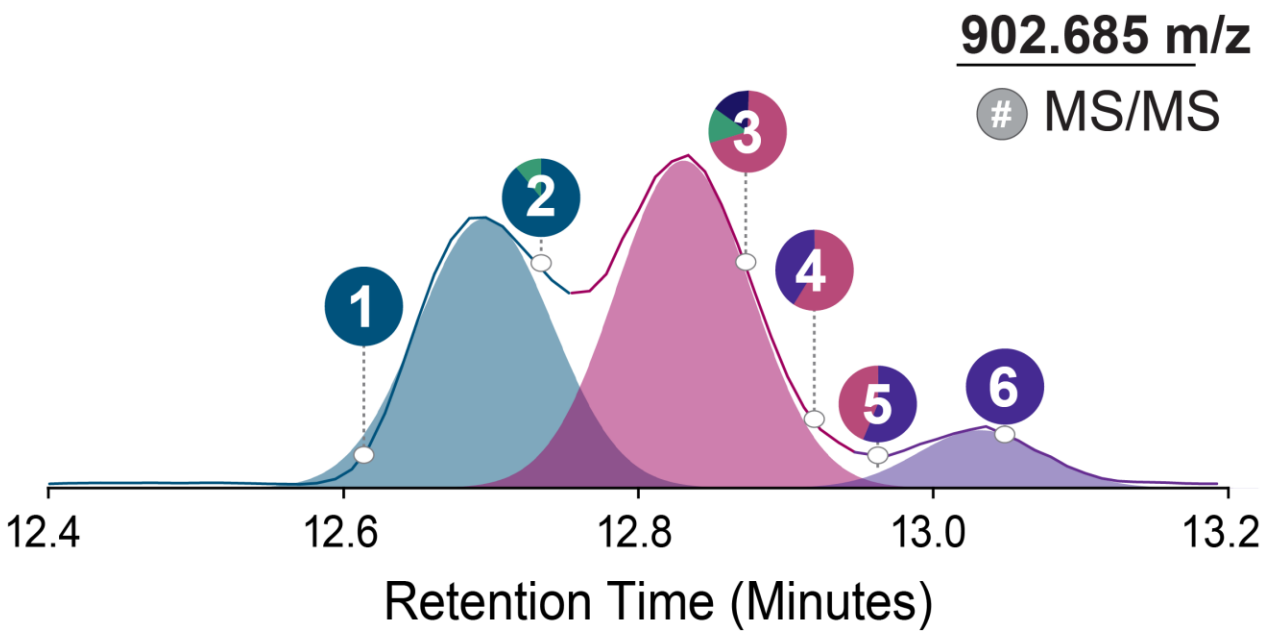
Lipid	Percent
PC 16:0_24:1	89%
PC 18:0_22:1	11%
PC 18:1_22:0	0%
PC 16:1_24:0	0%

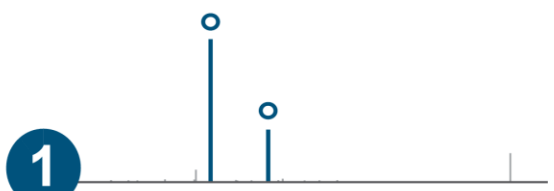


Spectral Purity

Lipid	Percent
PC 18:1_22:0	70%
PC 18:0_22:1	15%
PC 16:1_24:0	15%
PC 16:0_24:1	0%

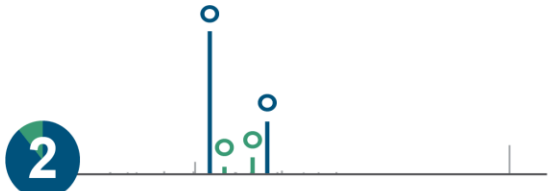
Isomers and Isobars are Key Sources of Complexity in Lipidomics





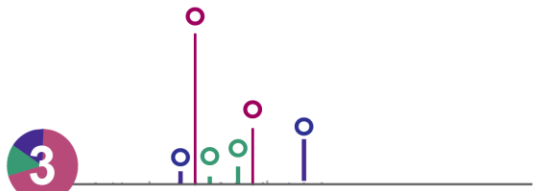
Spectral Purity

Lipid	Percent
PC 16:0_24:1	100%
PC 18:0_22:1	0%
PC 18:1_22:0	0%
PC 16:1_24:0	0%



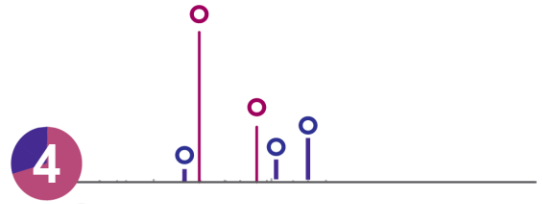
Spectral Purity

Lipid	Percent
PC 16:0_24:1	89%
PC 18:0_22:1	11%
PC 18:1_22:0	0%
PC 16:1_24:0	0%



Spectral Purity

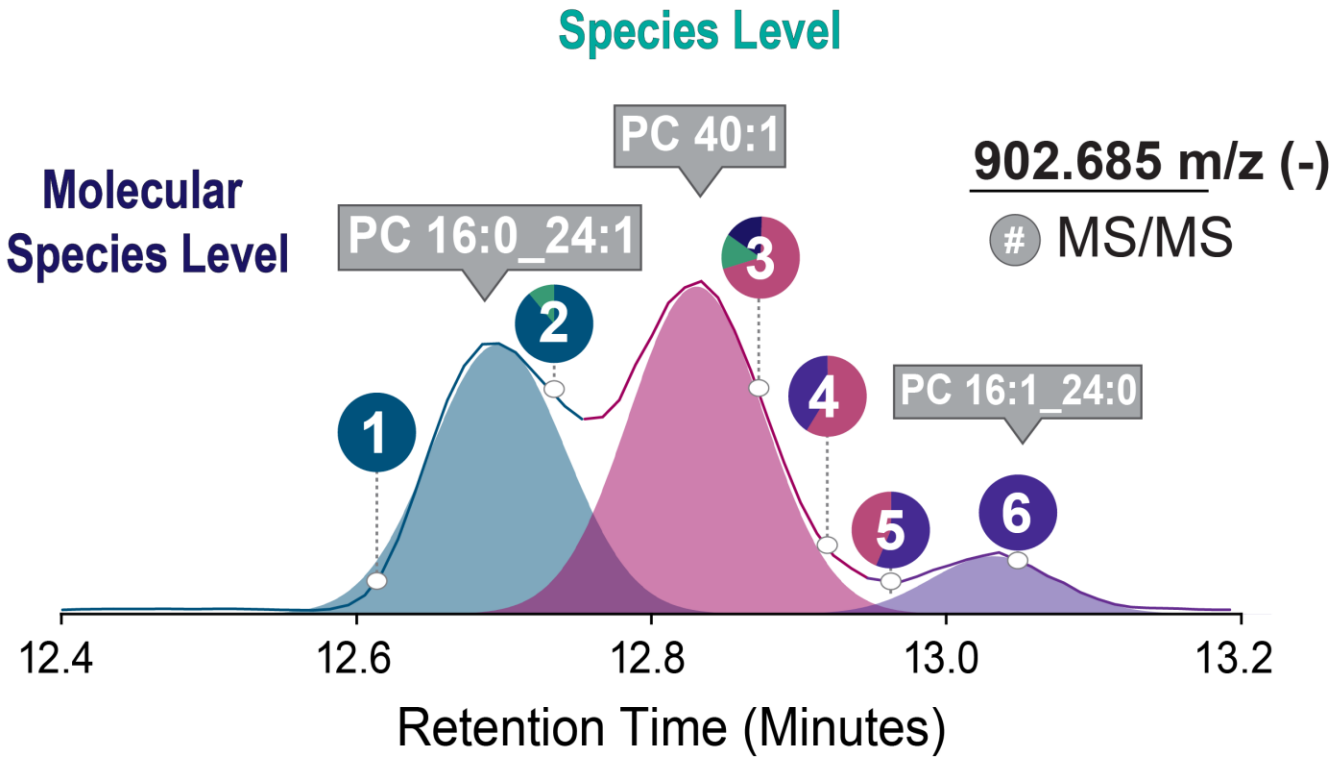
Lipid	Percent
PC 18:1_22:0	70%
PC 18:0_22:1	15%
PC 16:1_24:0	15%
PC 16:0_24:1	0%

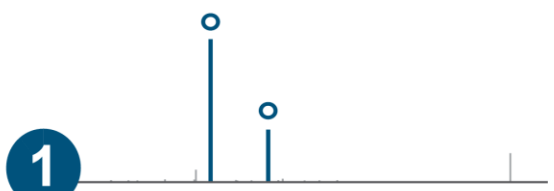


Spectral Purity

Lipid	Percent
PC 18:1_22:0	70%
PC 16:1_24:0	30%
PC 18:0_22:1	0%
PC 16:0_24:1	0%

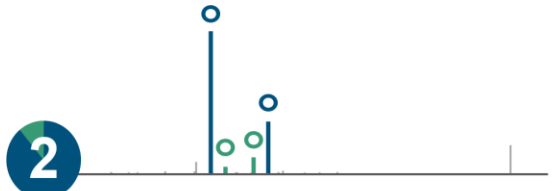
Isomers and Isobars are Key Sources of Complexity in Lipidomics





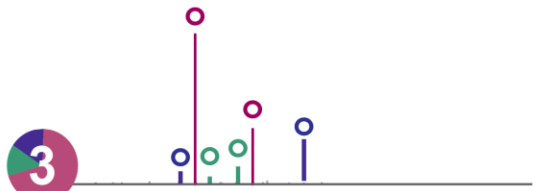
Spectral Purity

Lipid	Percent
PC 16:0_24:1	100%
PC 18:0_22:1	0%
PC 18:1_22:0	0%
PC 16:1_24:0	0%



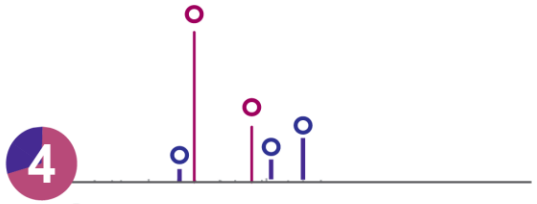
Spectral Purity

Lipid	Percent
PC 16:0_24:1	89%
PC 18:0_22:1	11%
PC 18:1_22:0	0%
PC 16:1_24:0	0%



Spectral Purity

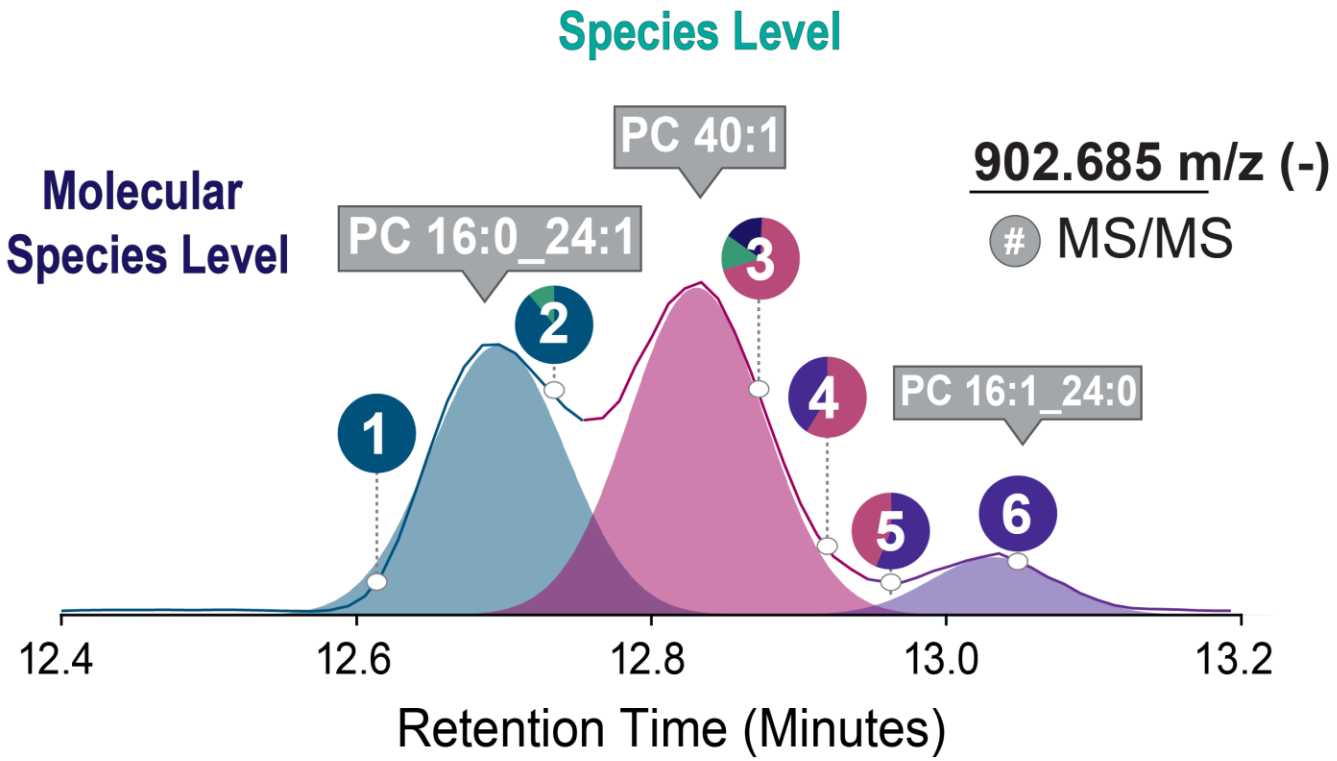
Lipid	Percent
PC 18:1_22:0	70%
PC 18:0_22:1	15%
PC 16:1_24:0	15%
PC 16:0_24:1	0%

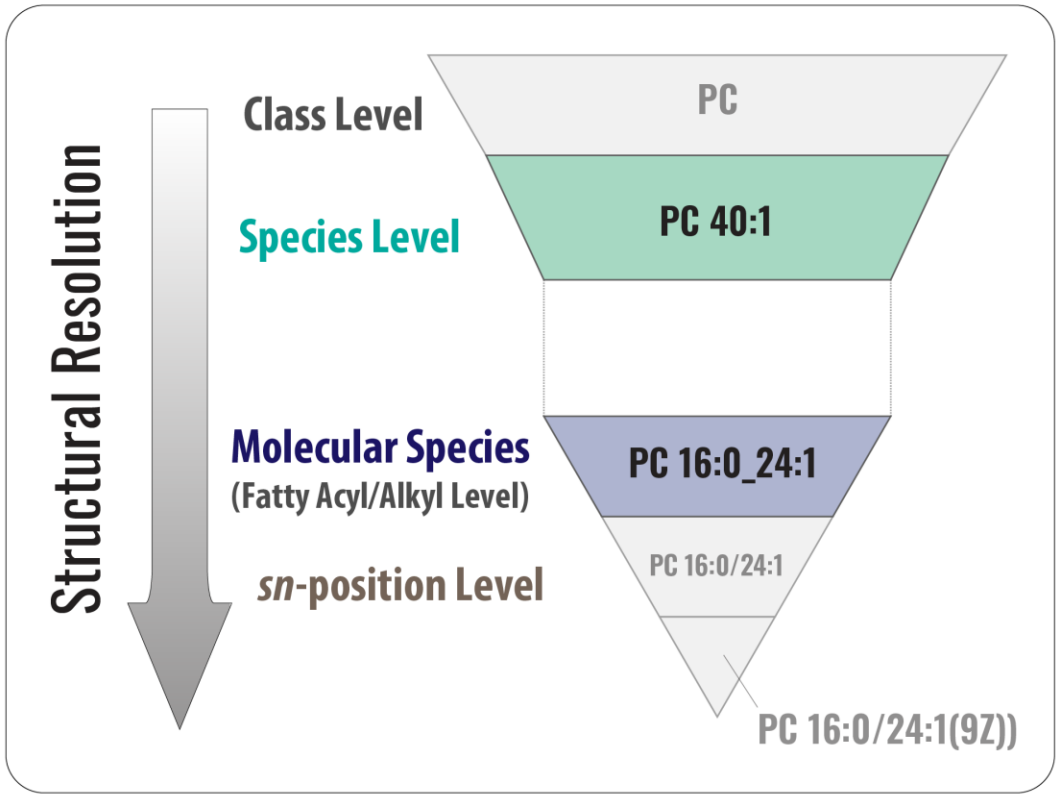


Spectral Purity

Lipid	Percent
PC 18:1_22:0	70%
PC 16:1_24:0	30%
PC 18:0_22:1	0%
PC 16:0_24:1	0%

Isomers and Isobars are Key Sources of Complexity in Lipidomics





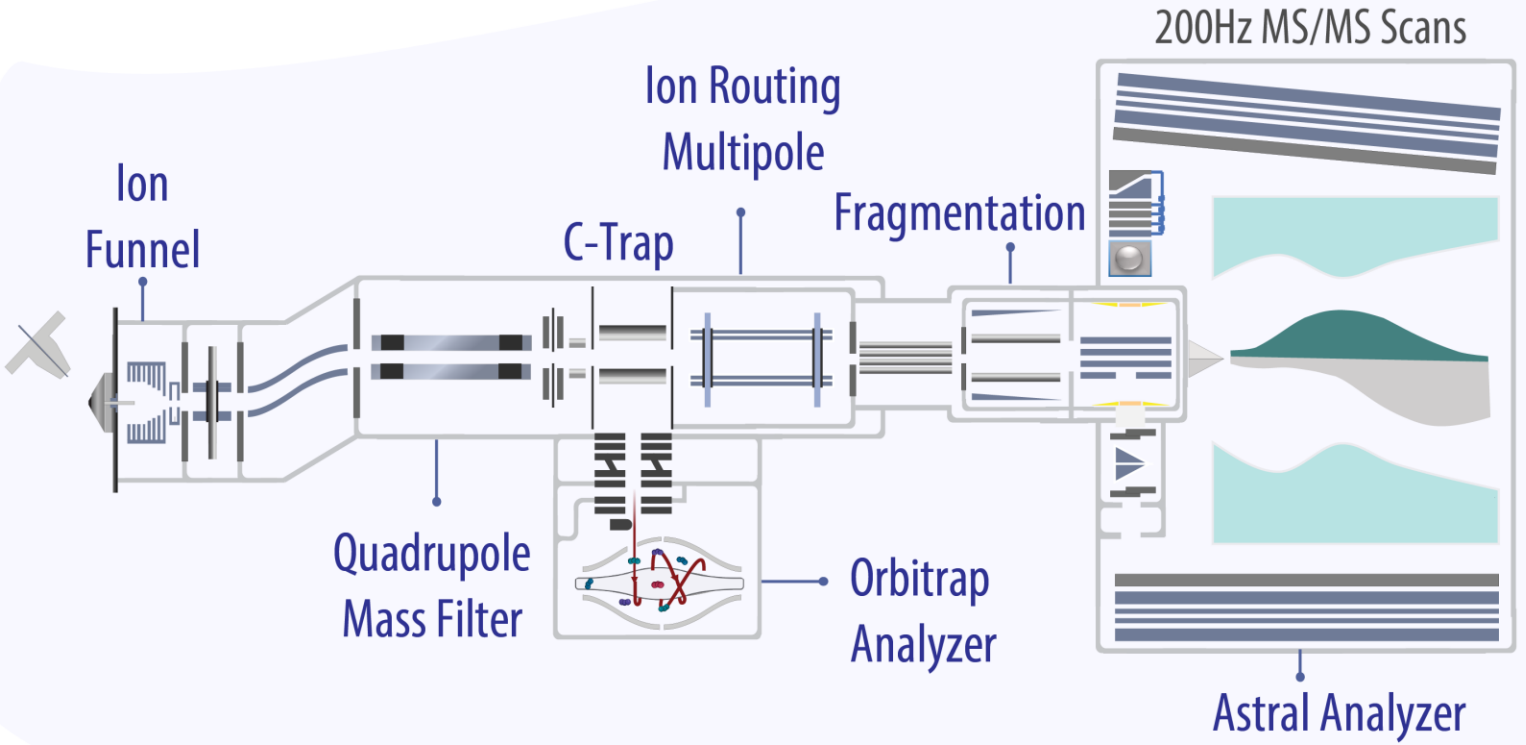
Randolph, C. et. al., Trends in Analytical Chemistry (2023)

Orbitrap Astral as a Promising Platform to Untangle Lipidomic Complexity

High-speed, high-sensitivity MS/MS with lossless ion transfer, enabling fast analytical acquisition rates

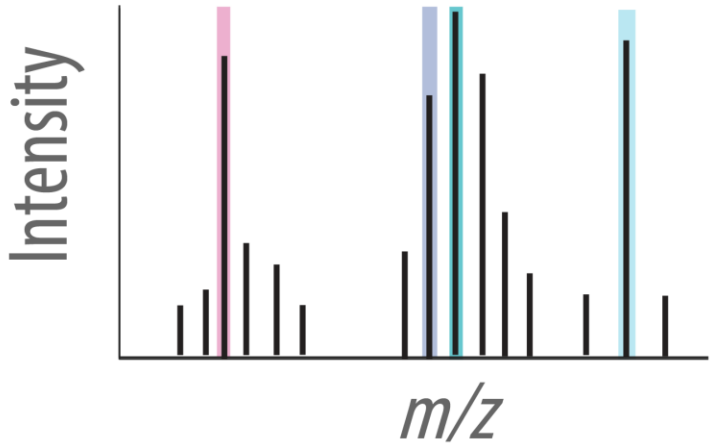


Orbitrap Astral MS

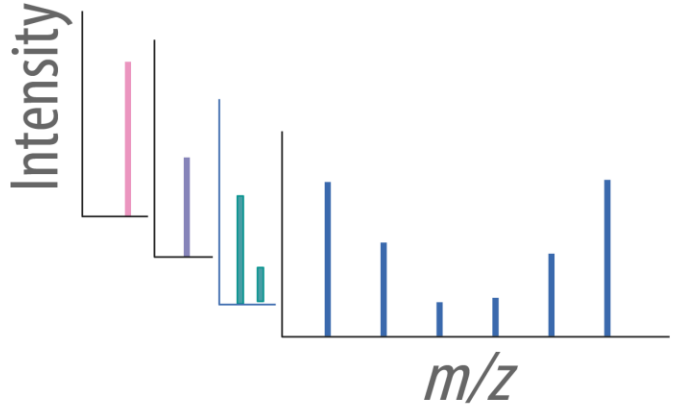


Data Acquisition Strategies on the Orbitrap Astral MS

Data Dependent Acquisition (DDA)

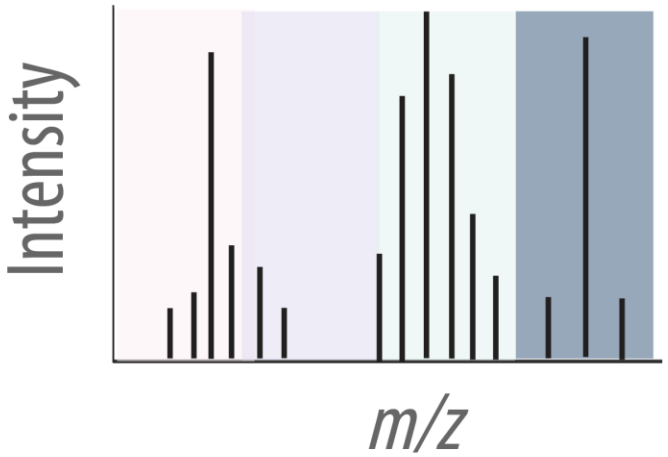


Precursor Selection

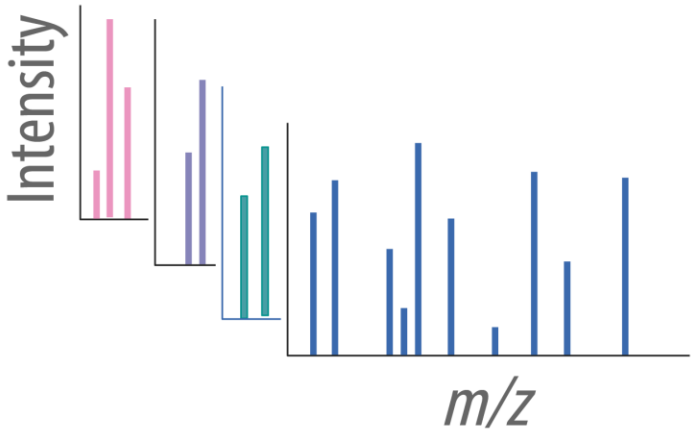


Fragmentation of Selected Precursors

Data Independent Acquisition (DIA)



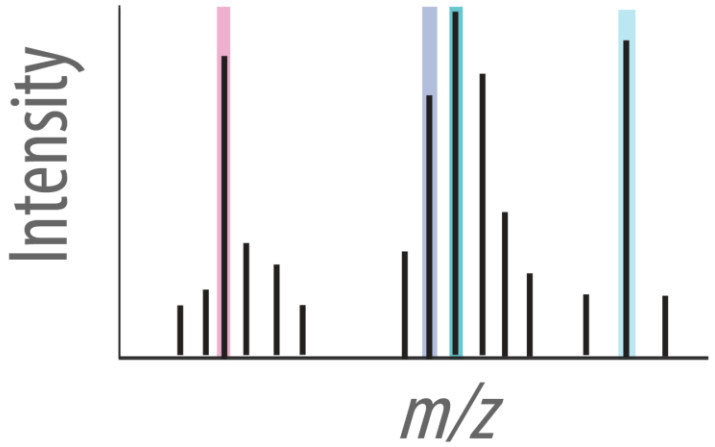
Scan Across All Isolation Windows



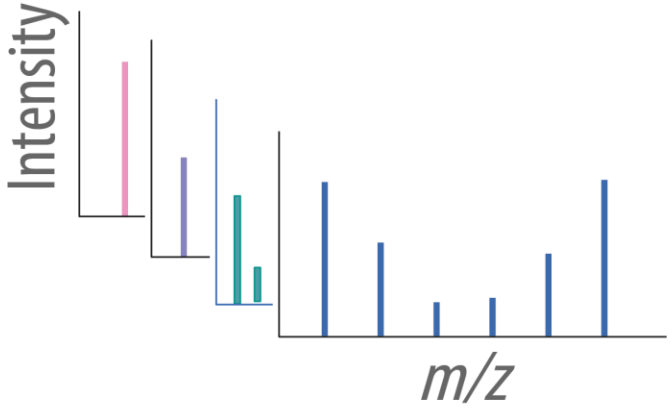
Fragmentation of All Precursors in each window

Data Acquisition Strategies on the Orbitrap Astral MS

Data Dependent Acquisition (DDA)

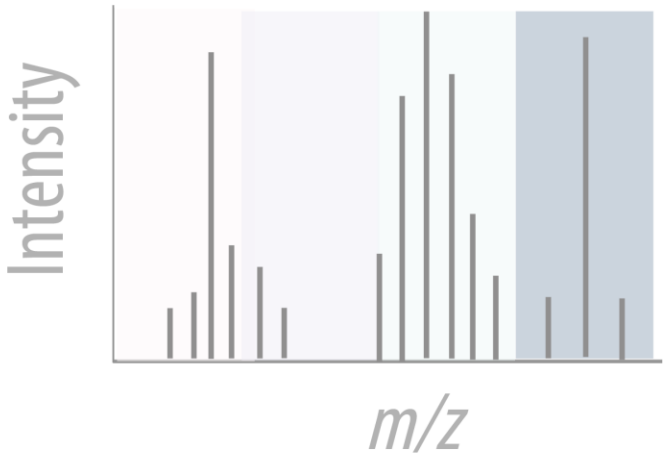


Precursor Selection

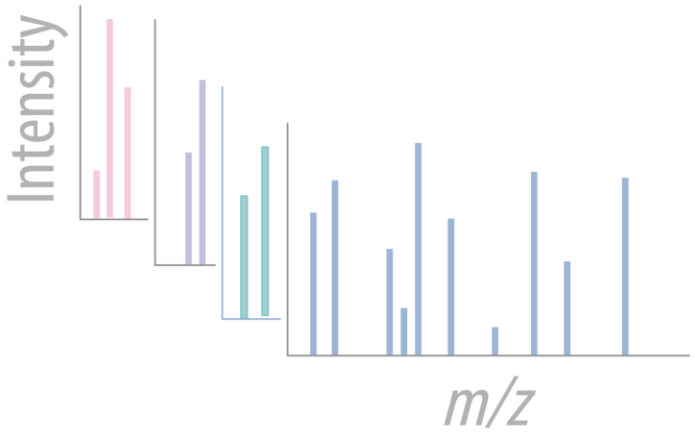


Fragmentation of Selected Precursors

Data Independent Acquisition (DIA)

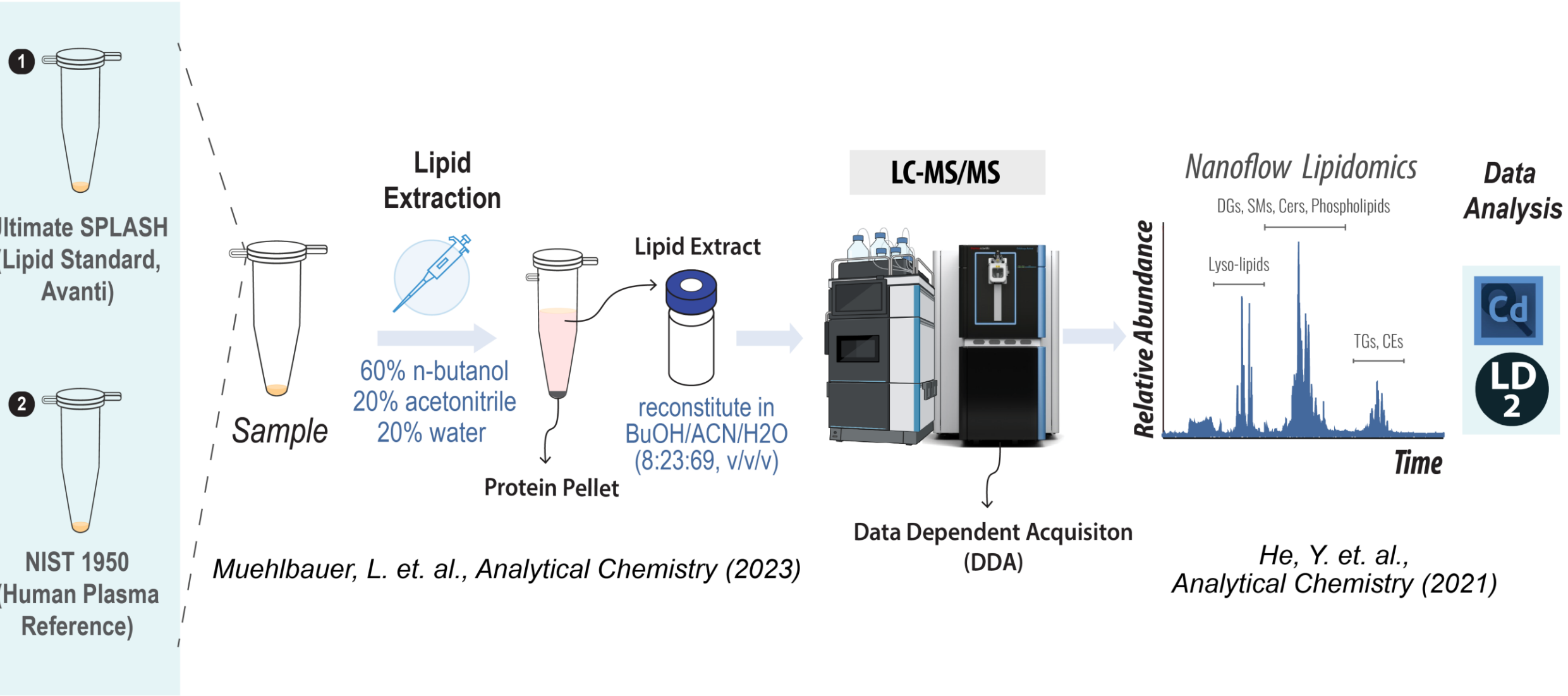


Scan Across All Isolation Windows



Fragmentation of All Precursors in each window

Experimental Workflow

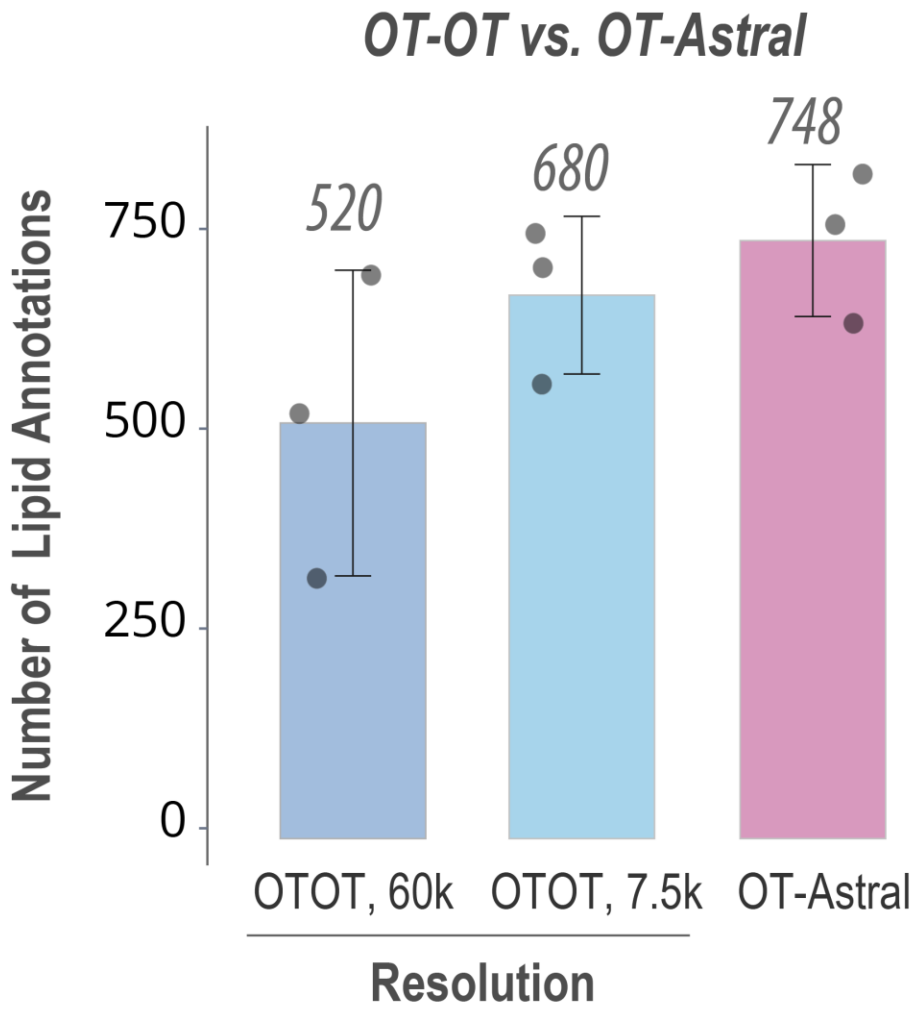


Muehlbauer, L. et. al., Analytical Chemistry (2023)

He, Y. et. al., Analytical Chemistry (2021)

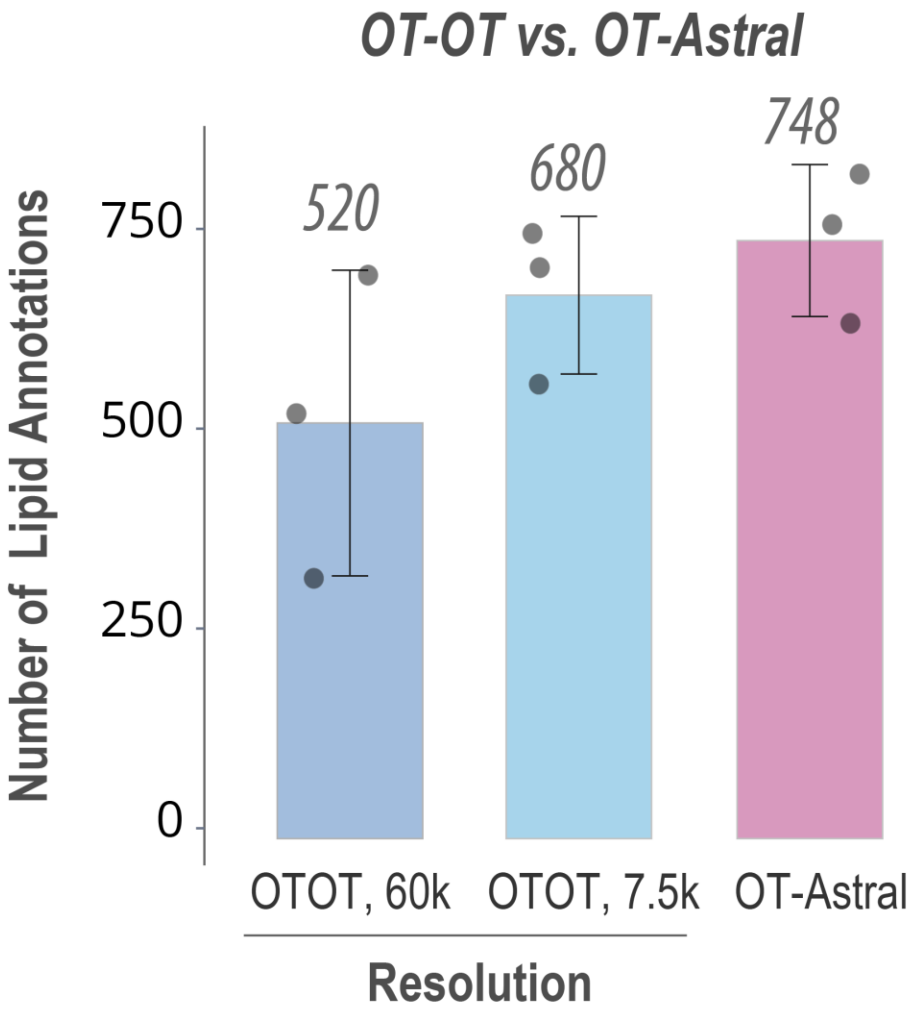
Increased Number of MS2 Scans Does Not Translate into Increased Lipid IDs

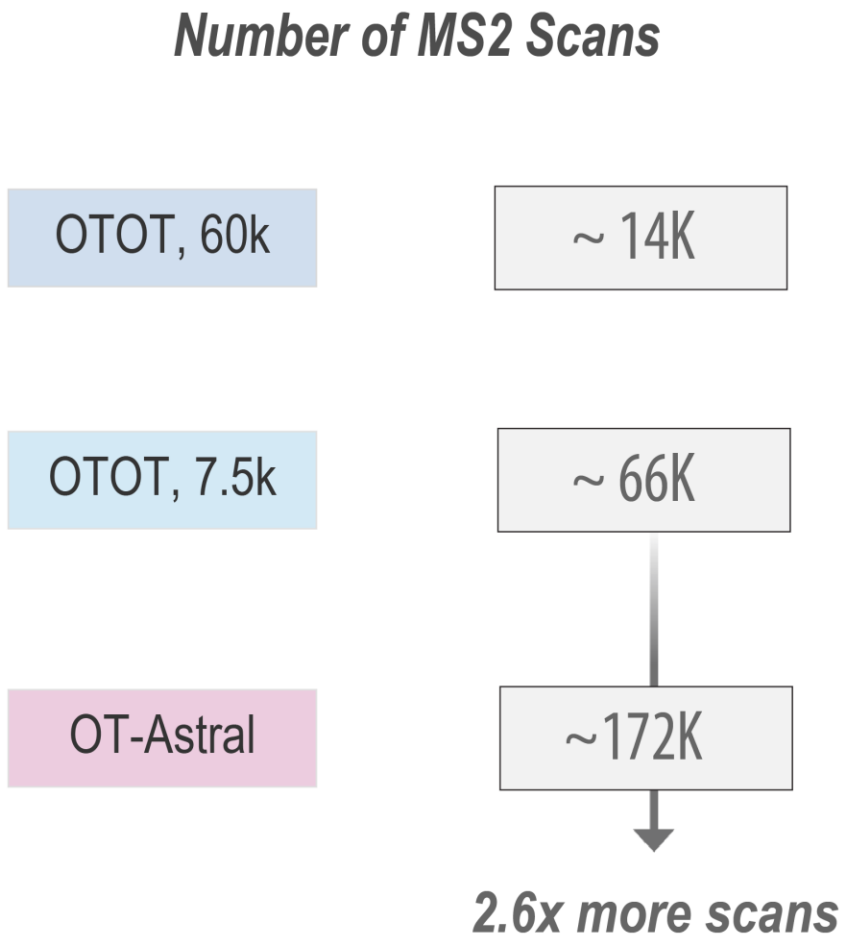
NIST 1950 Human Plasma



Increased Number of MS2 Scans Does Not Translate into Increased Lipid IDs

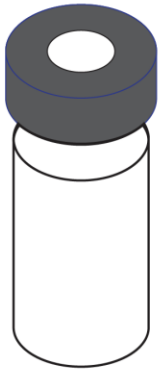
NIST 1950 Human Plasma



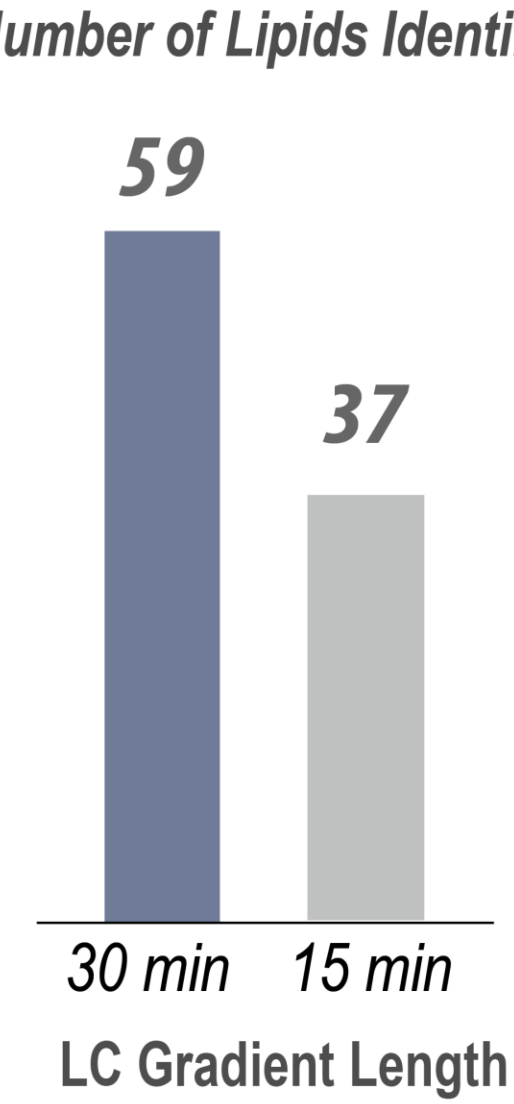


Increased Number of MS2 Scans Does Not Translate into Increased Lipid IDs

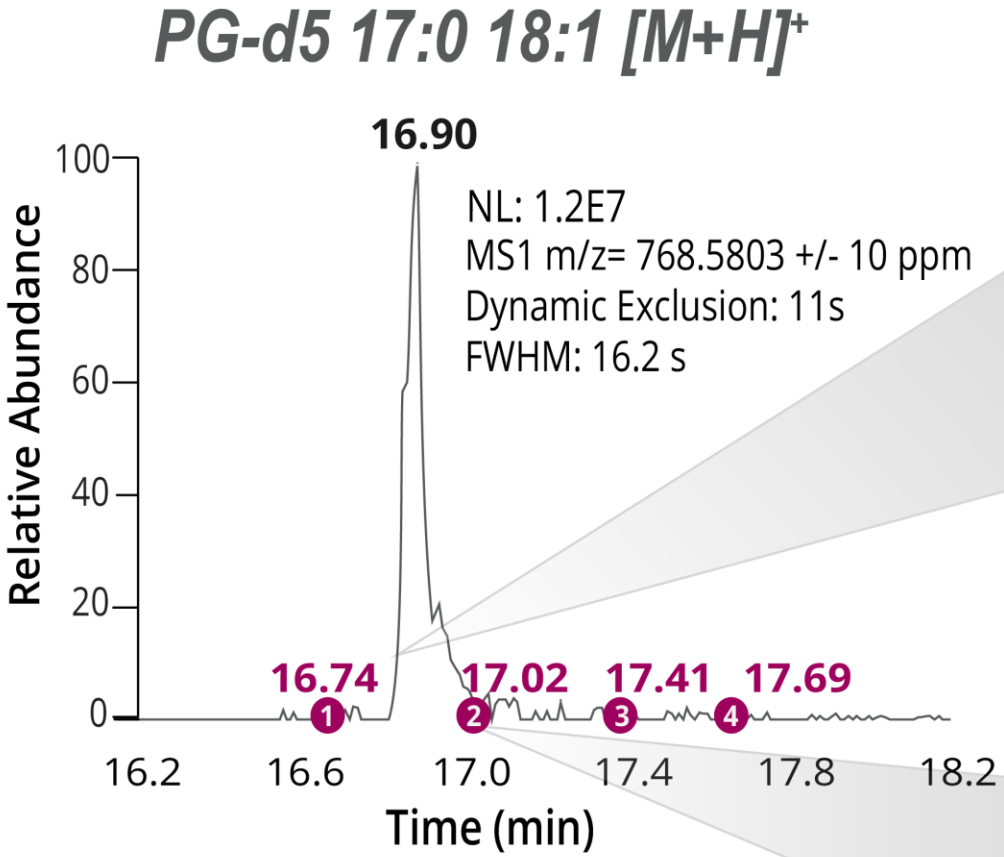
UltimateSPLASH Lipid Standard



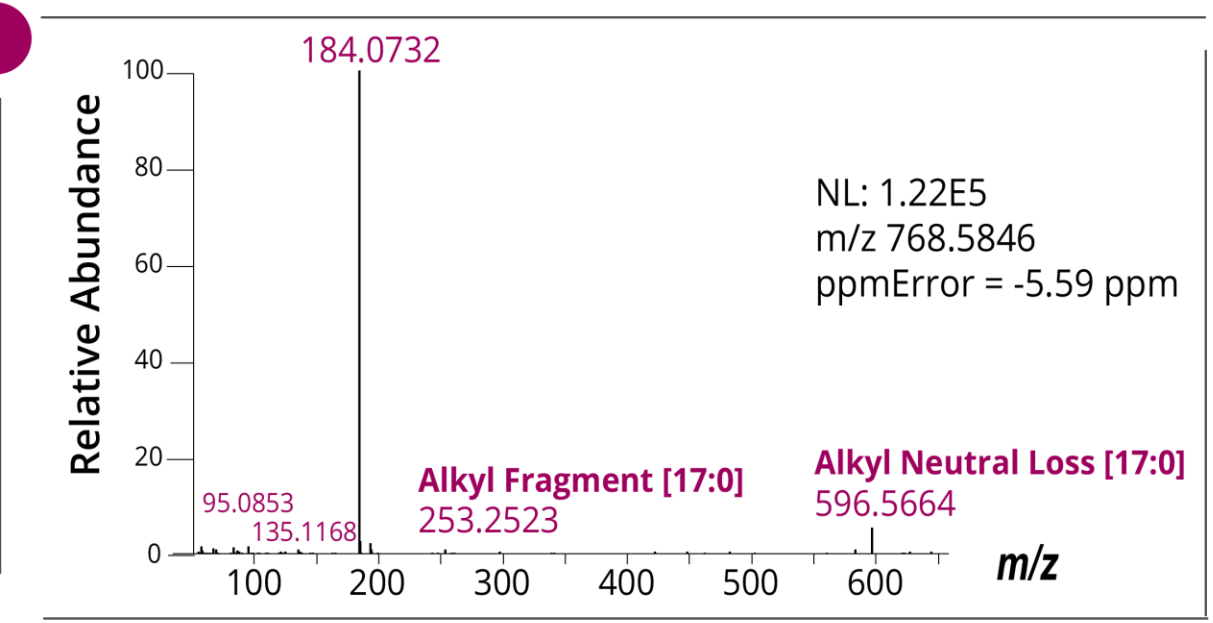
Lipid Class	Count
CE-d7	5
Cer-d7	5
DG-d5	5
LysoPC-d5	3
LysoPE-d5	3
LysoPG-d5	3
LysoPI-d5	3
LysoPS-d5	3
PC-d5	5
PE-d5	5
PG-d5	5
PI-d5	5
PS-d5	5
SM-d9	5
TG-d5	9
Total	**69**

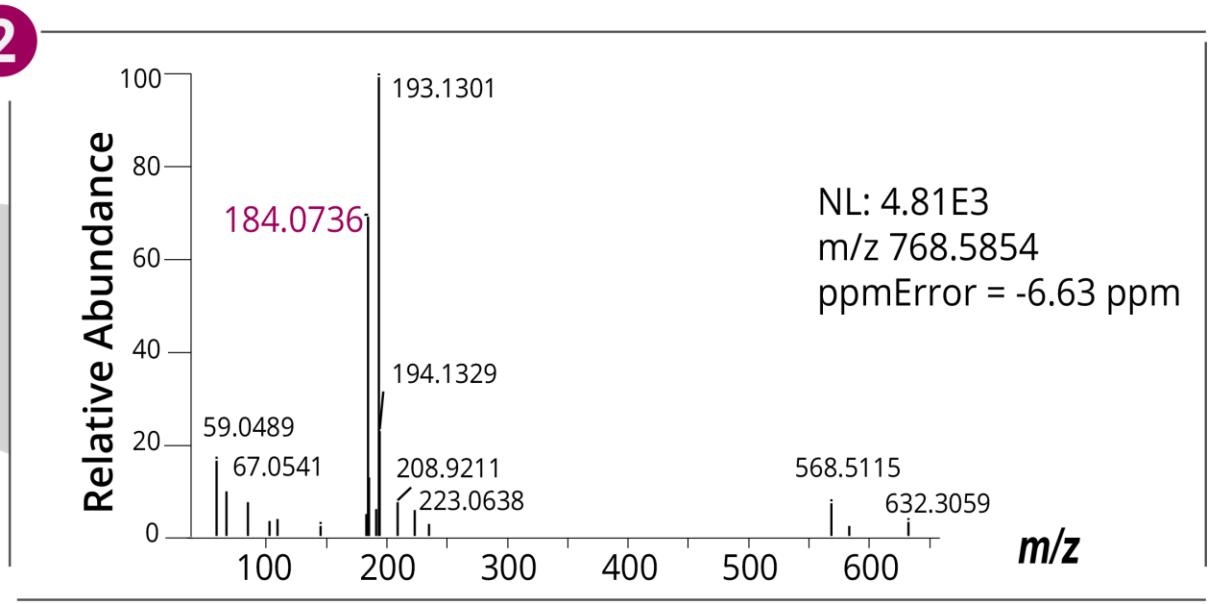


Poorly Timed DDA MS/MS Sampling Reduces Spectral Quality



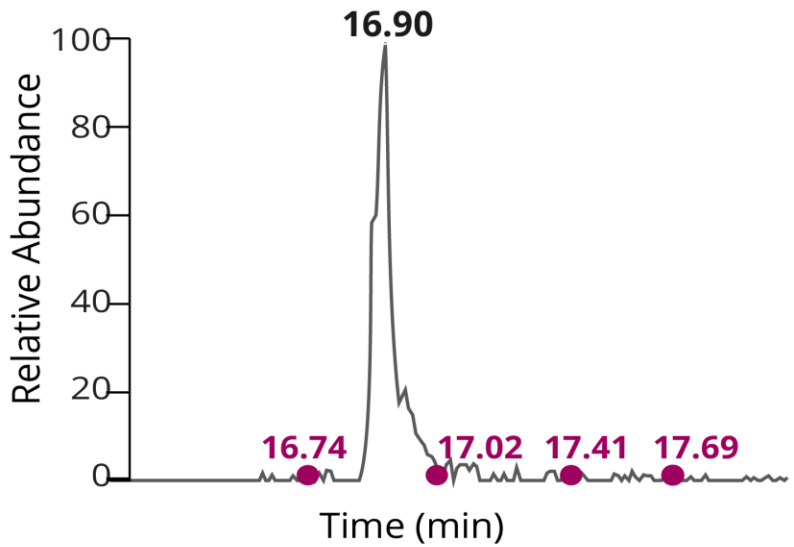
UltimateSPLASH
DDA Positive mode
0.6s Cycle time



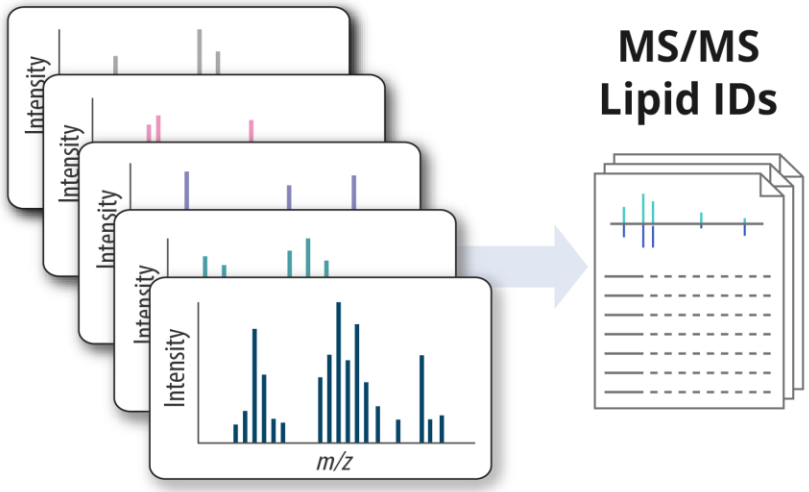


Low Number of Lipid Identifications

Sampling issues

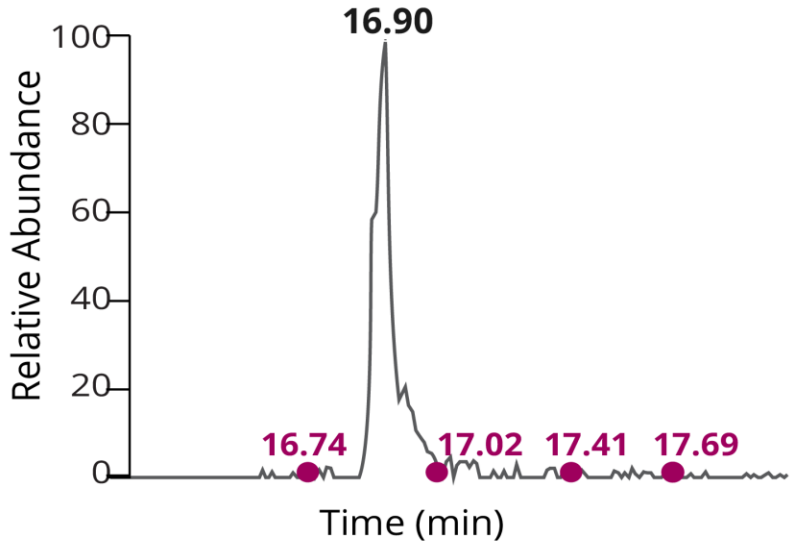


Turning Scans into IDs

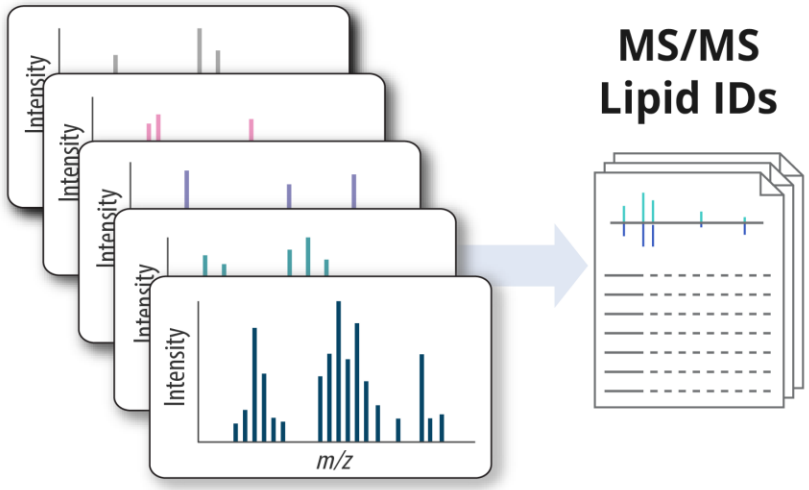


Low Number of Lipid Identifications

Sampling issues

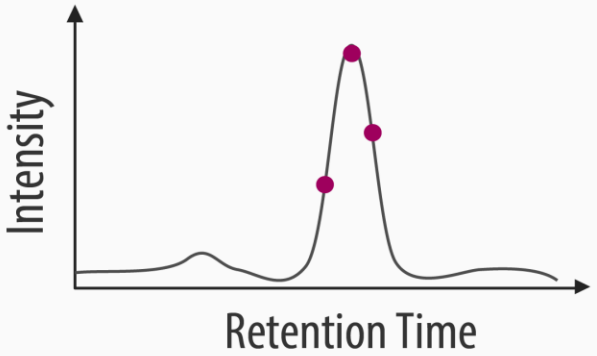


Turning Scans into IDs

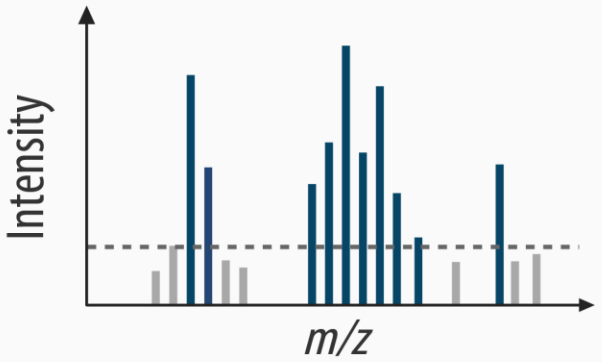


MS/MS Parameters to Optimize

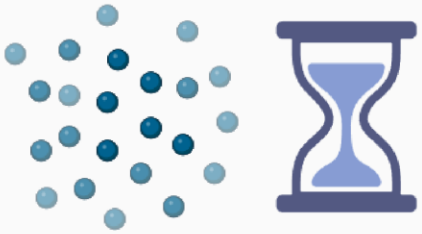
Dynamic Exclusion



Intensity Threshold

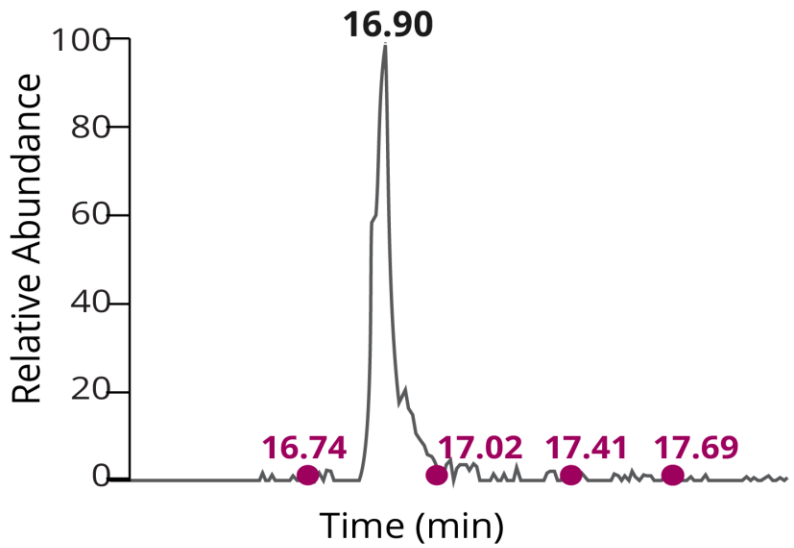


Automatic Gain Control and Max Injection Time

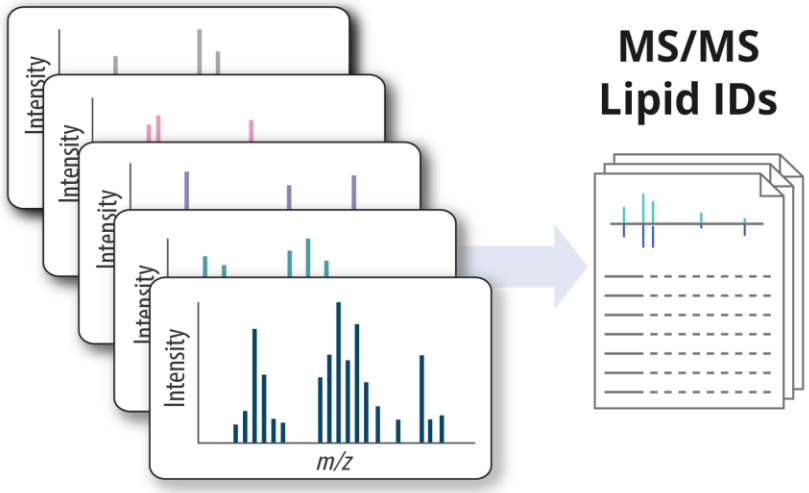


Low Number of Lipid Identifications

Sampling issues

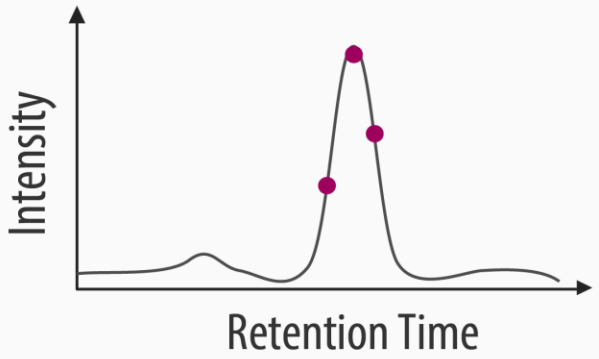


Turning Scans into IDs

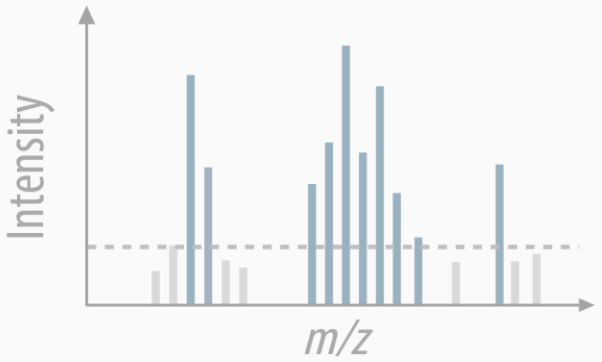


MS/MS Parameters to Optimize

Dynamic Exclusion



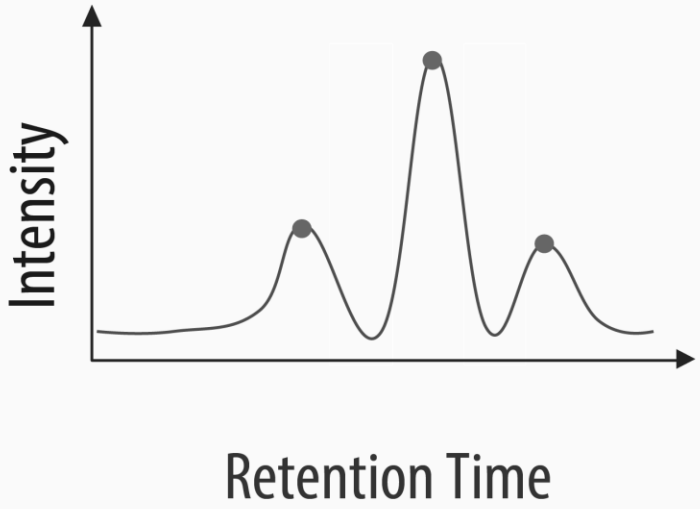
Intensity Threshold

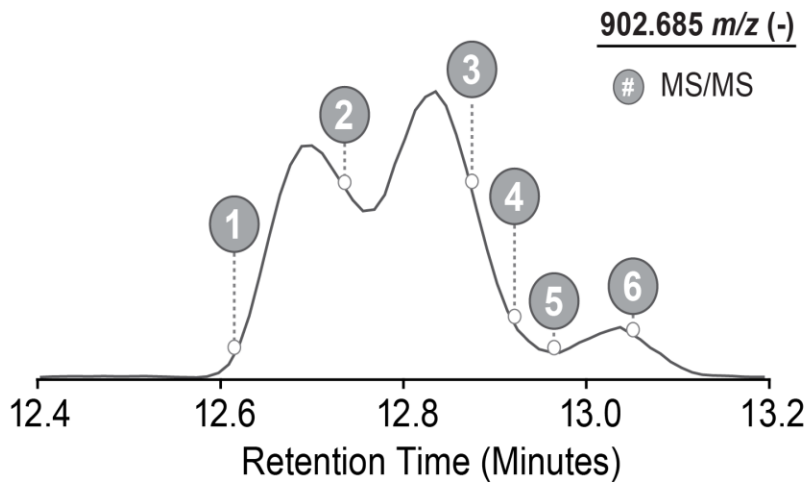


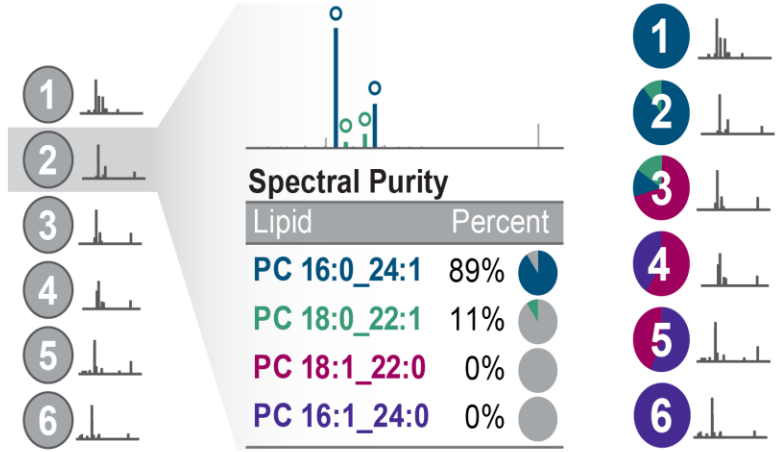
Automatic Gain Control and Max Injection Time

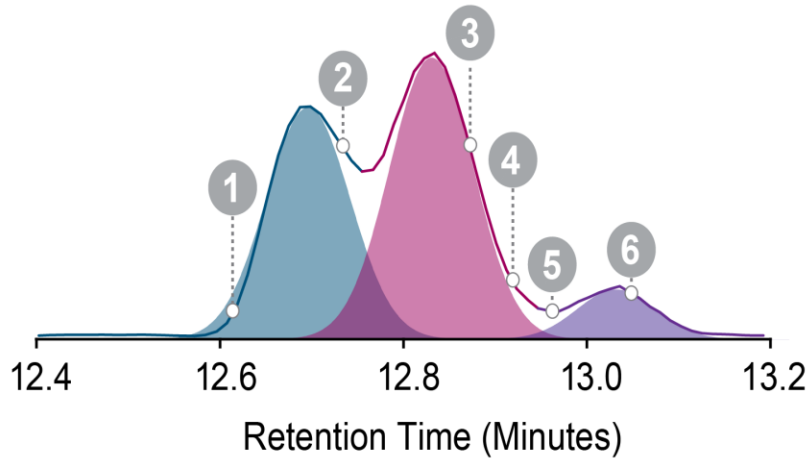


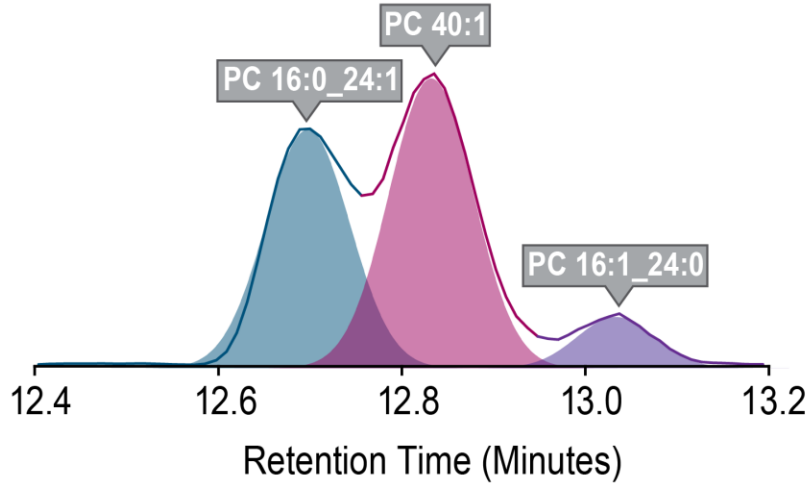
Dynamic Exclusion





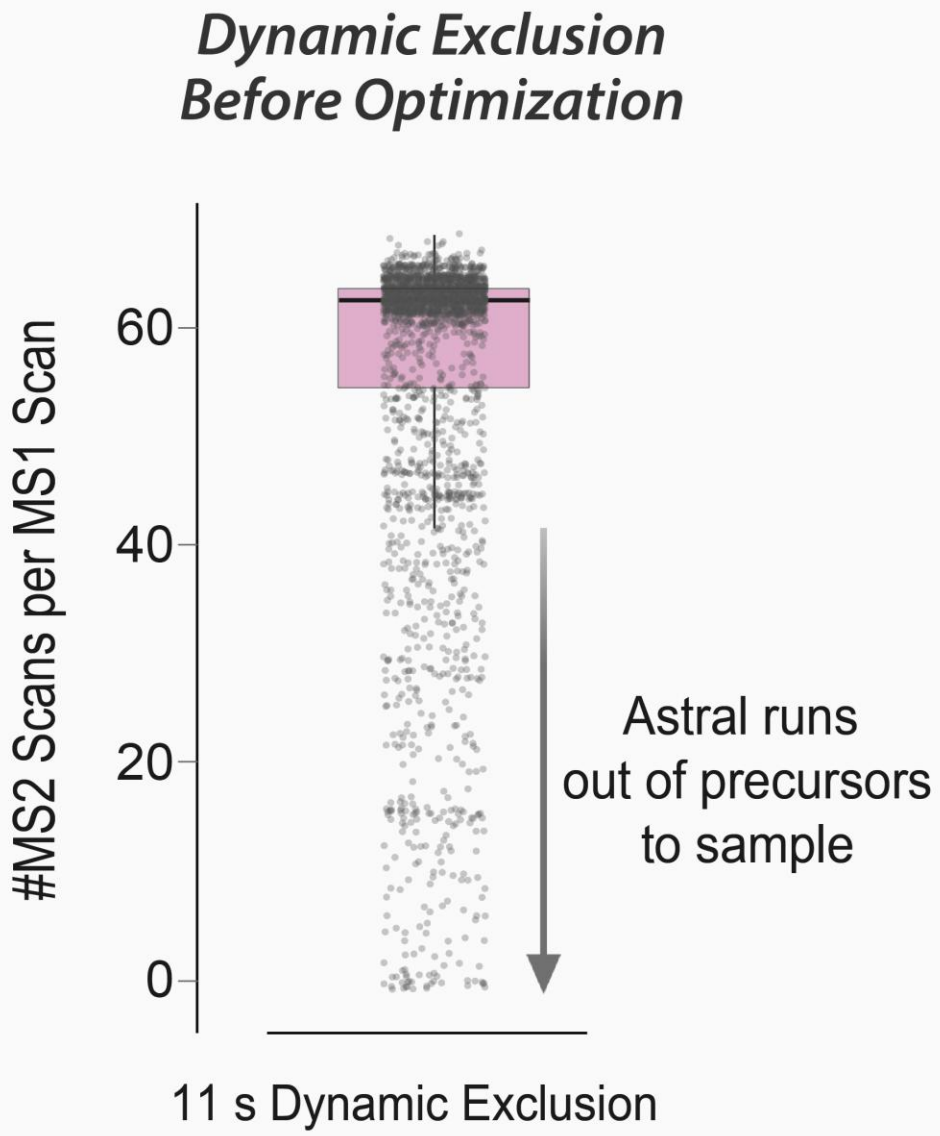




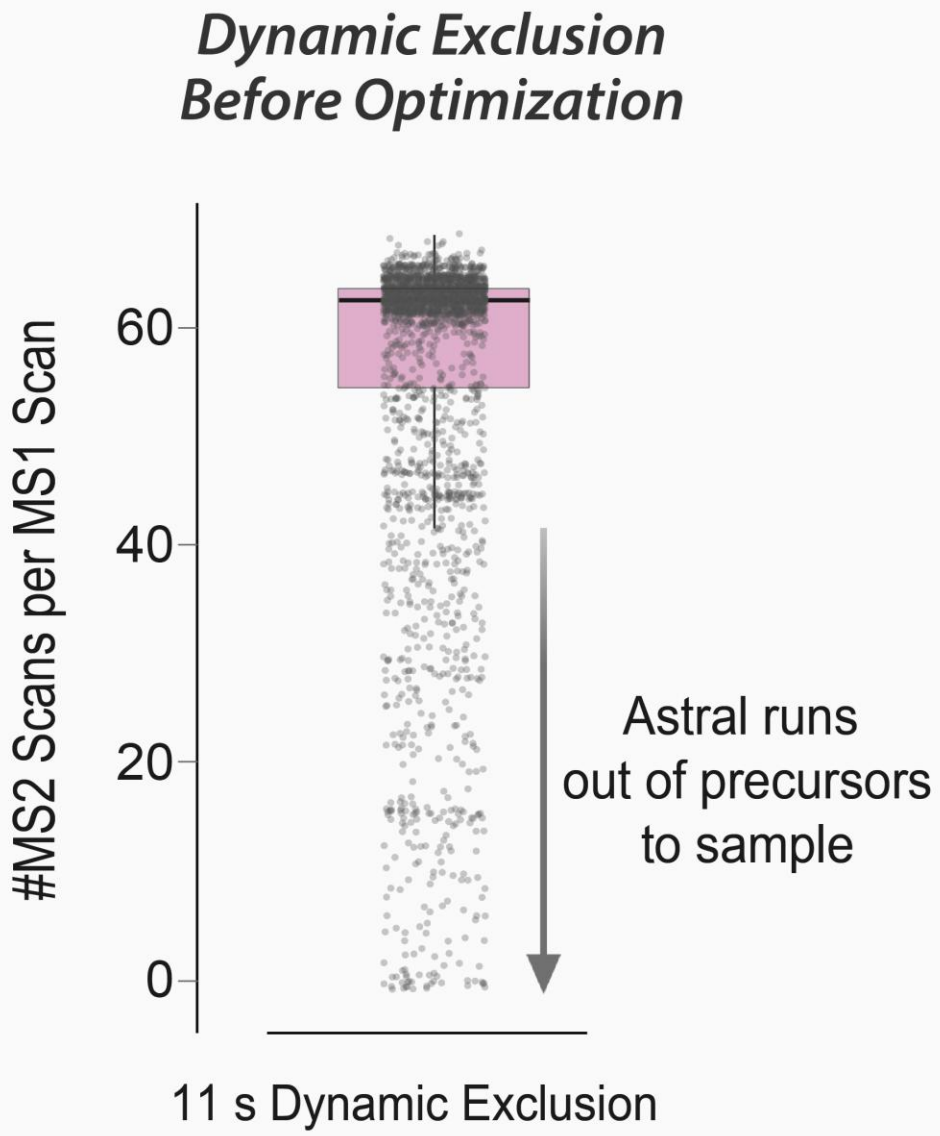


Hutchins, P. et. al., Cell Systems (2019)

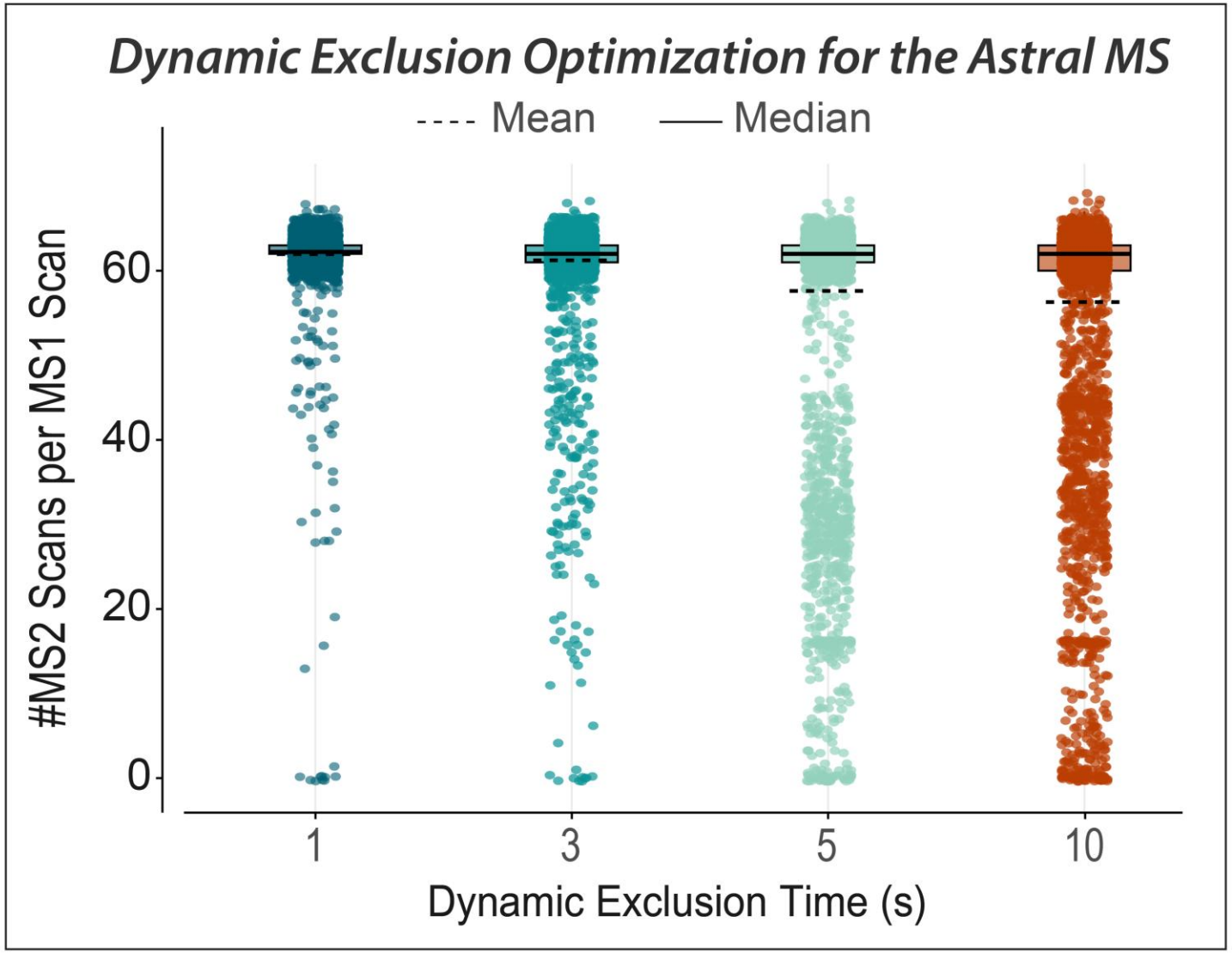
Astral's Speed Requires Shorter Dynamic Exclusion Times for Better Sampling



Astral's Speed Requires Shorter Dynamic Exclusion Times for Better Sampling



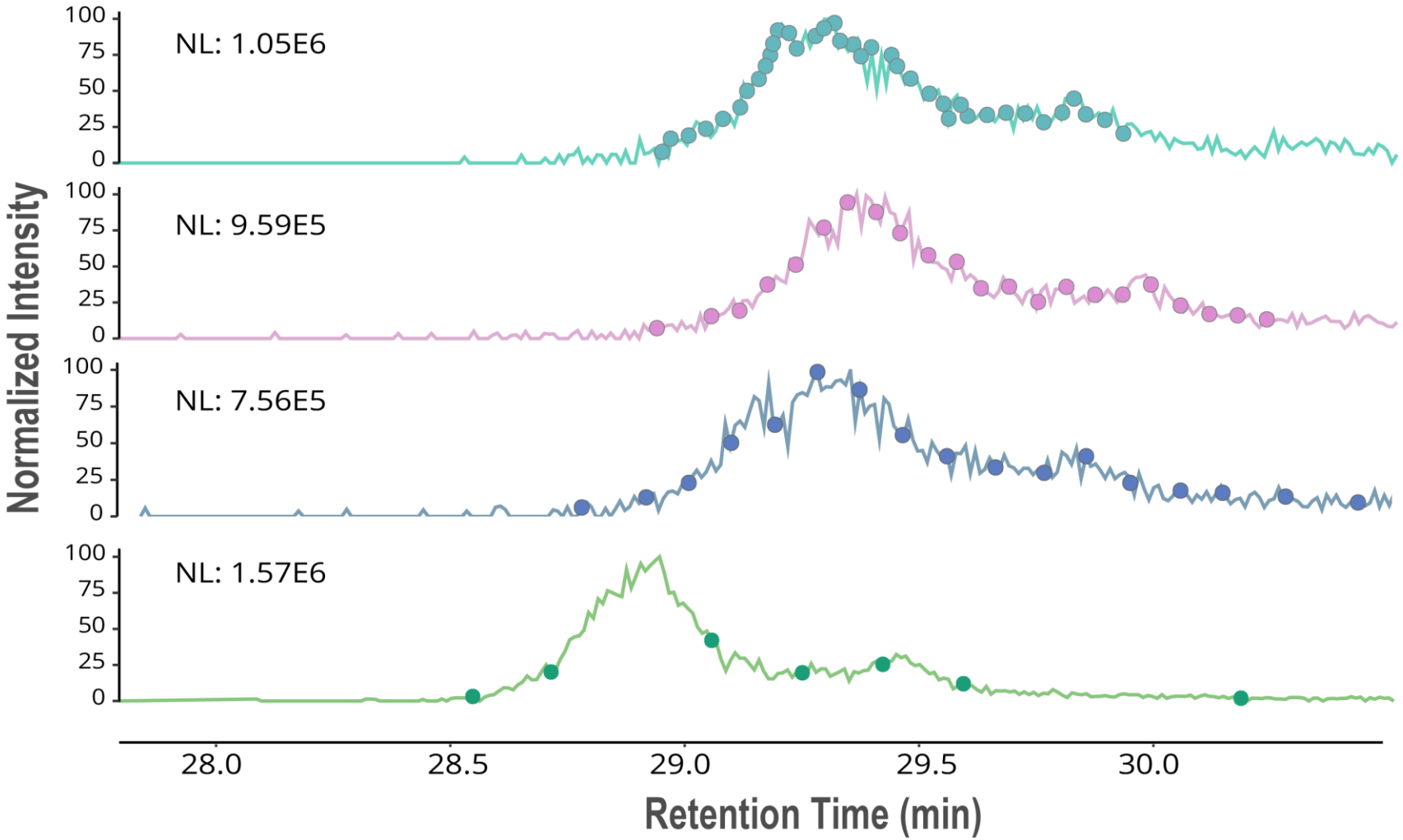
NIST 1950 Human Plasma



Shorter Dynamic Exclusion Times Increase MS2 Scans Across the Peak

TG 12:0_12:0_18:2 [M +NH$_4$]$^+$

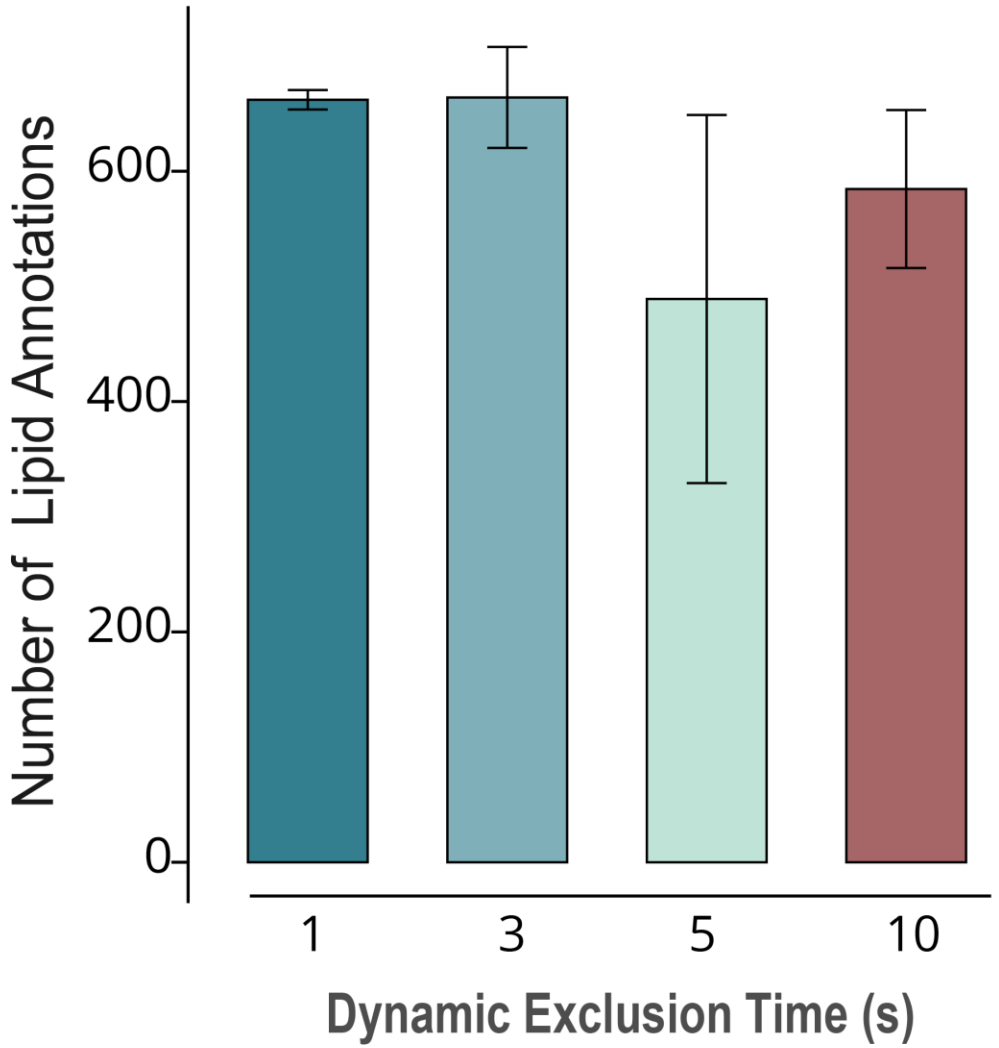
m/z = 736.6475 +/- 5ppm



Dynamic Exclusion	Identified
1s MS2 Scans = 39	✓
3s MS2 Scans = 24	✓
5s MS2 Scans = 22	✗
10s MS2 Scans = 9	✗

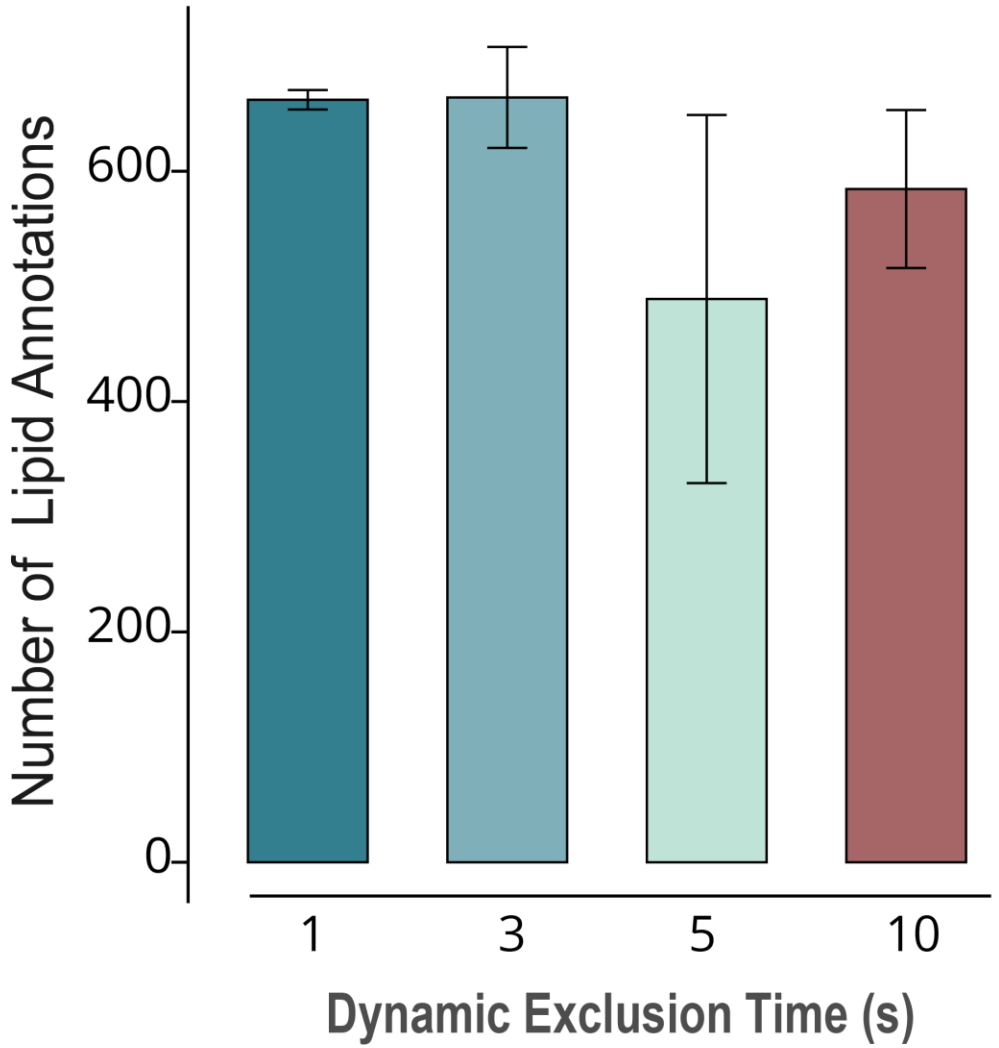
Shorter Dynamic Exclusion Times Improve Lipid IDs But Not Spectral Purity

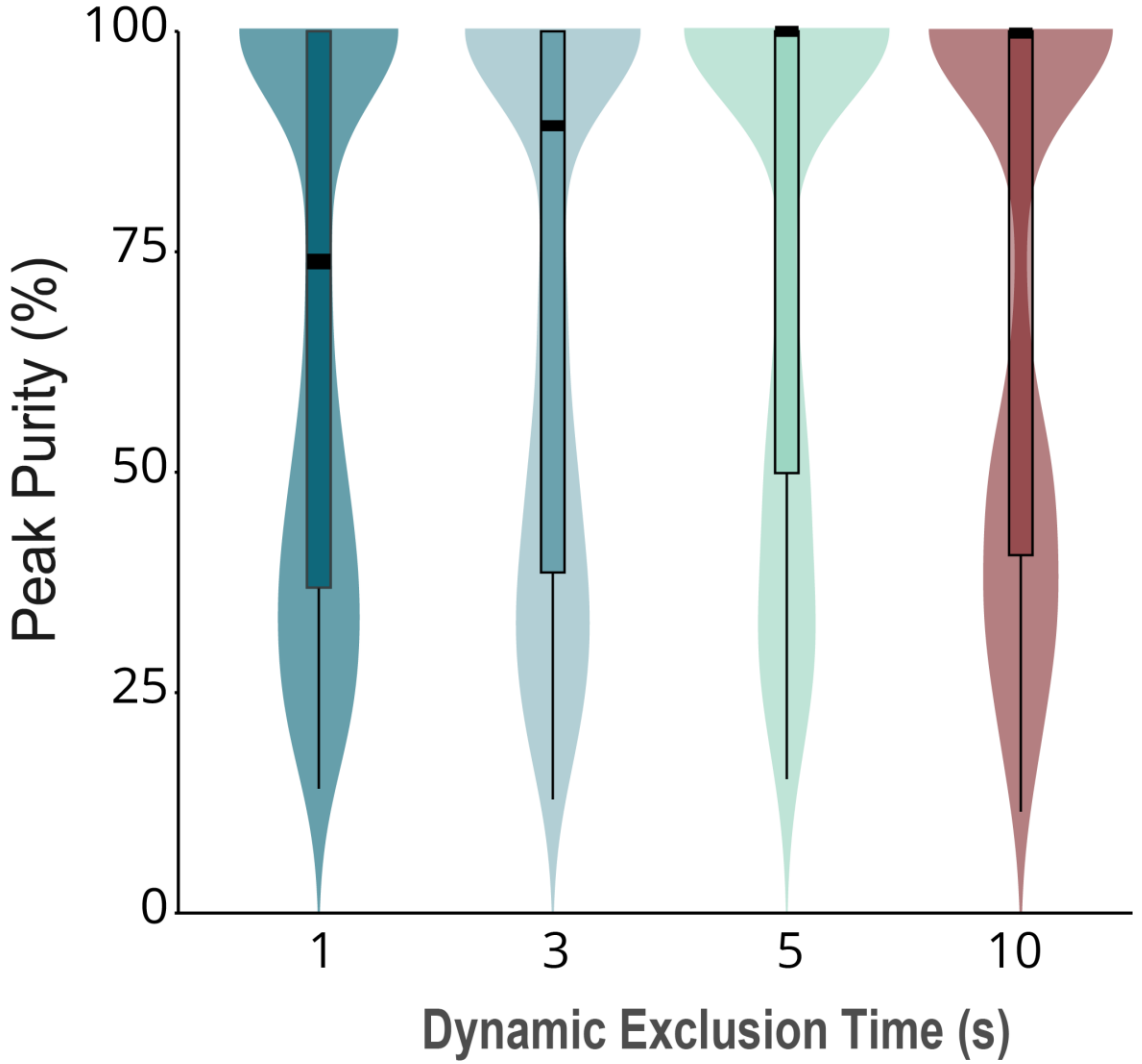
Dynamic Exclusion



Shorter Dynamic Exclusion Times Improve Lipid IDs But Not Spectral Purity

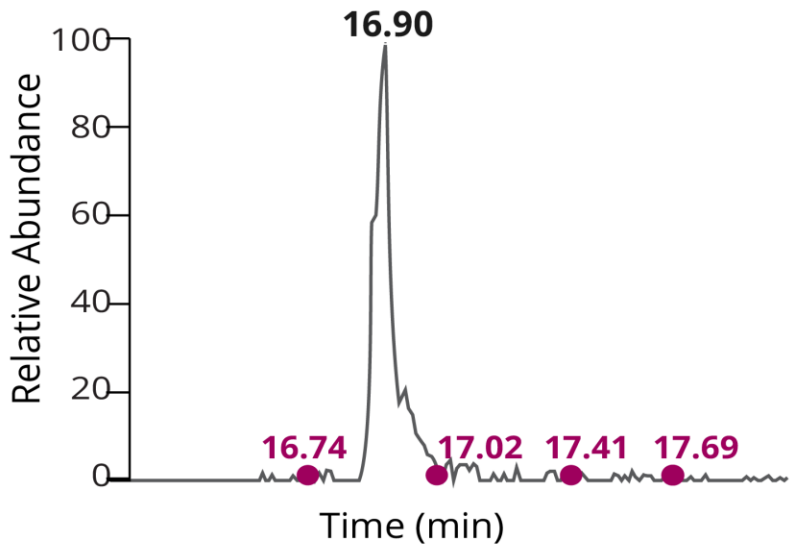
Dynamic Exclusion



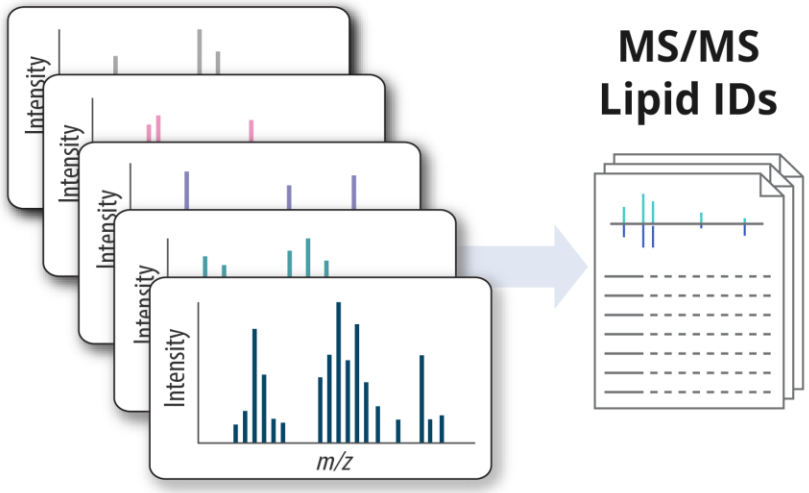


Low Number of Lipid Identifications

Sampling issues



Turning Scans into IDs

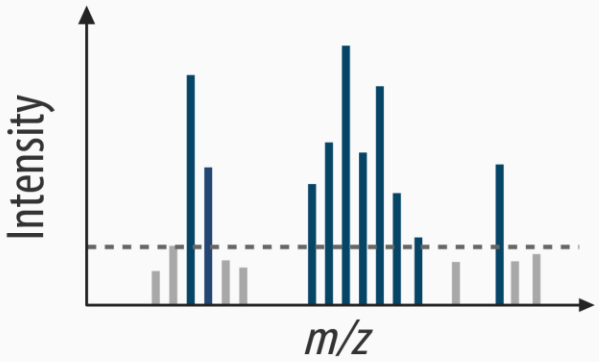


MS/MS Parameters to Optimize

Dynamic Exclusion



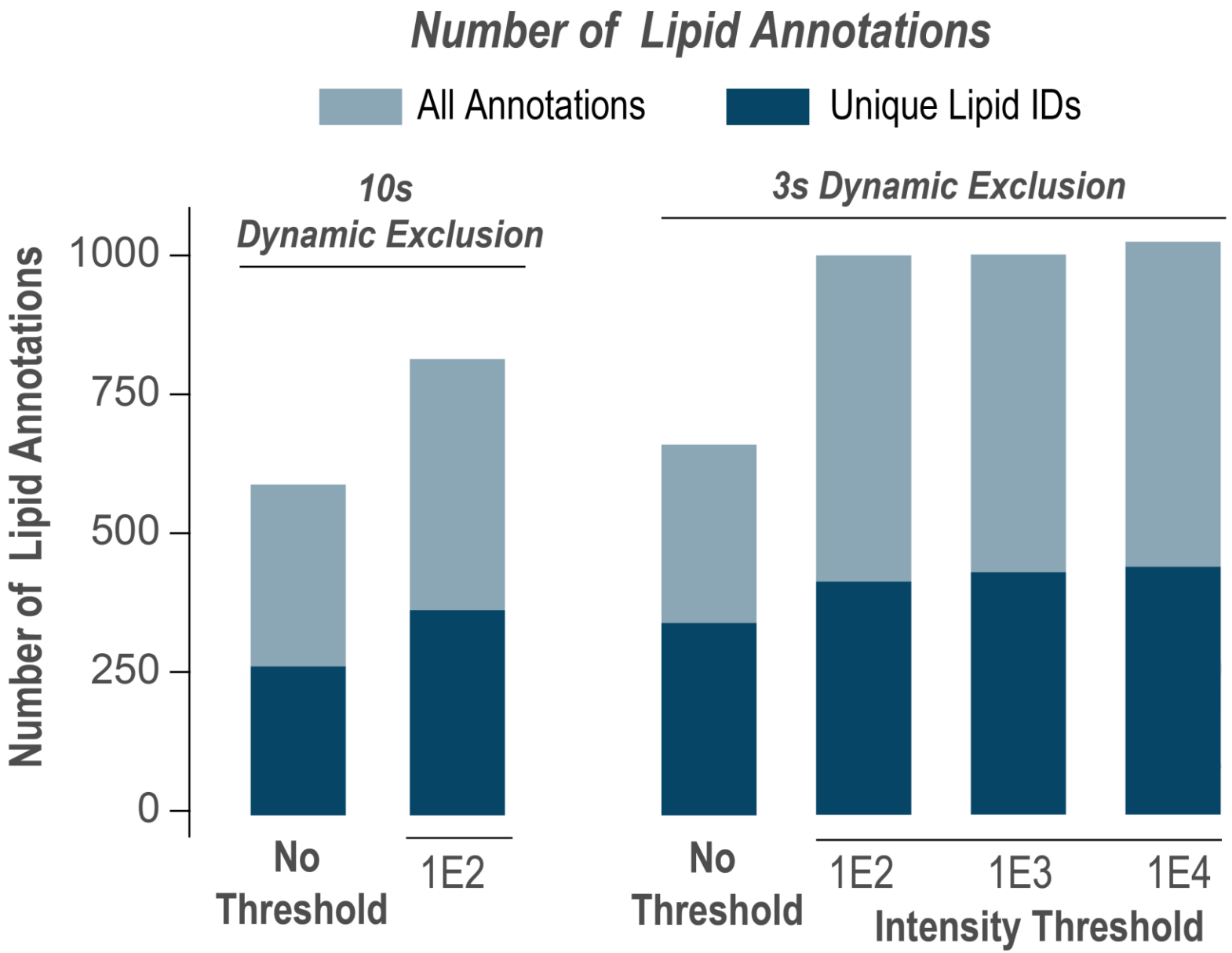
Intensity Threshold



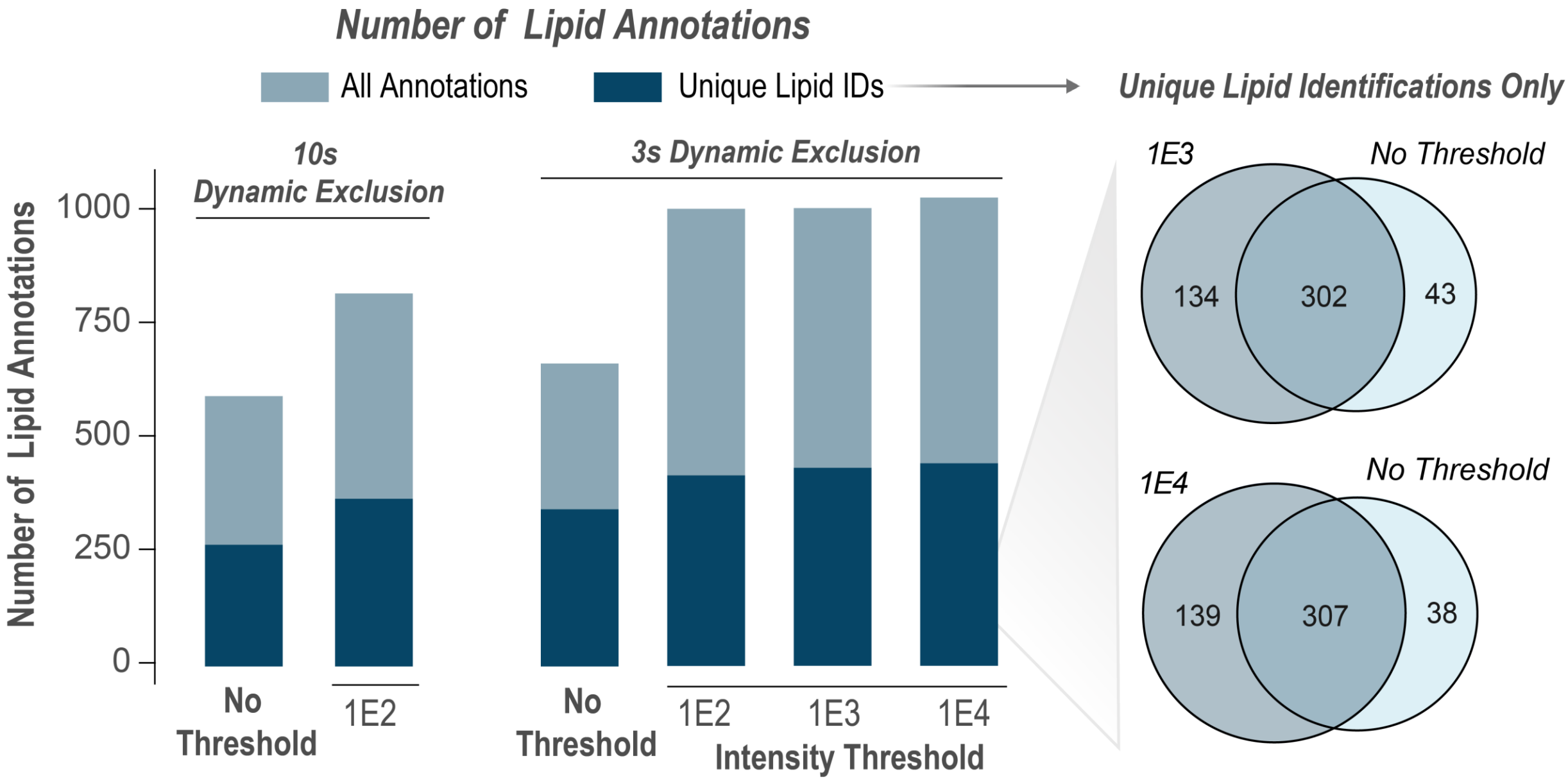
Automatic Gain Control and Max Injection Time



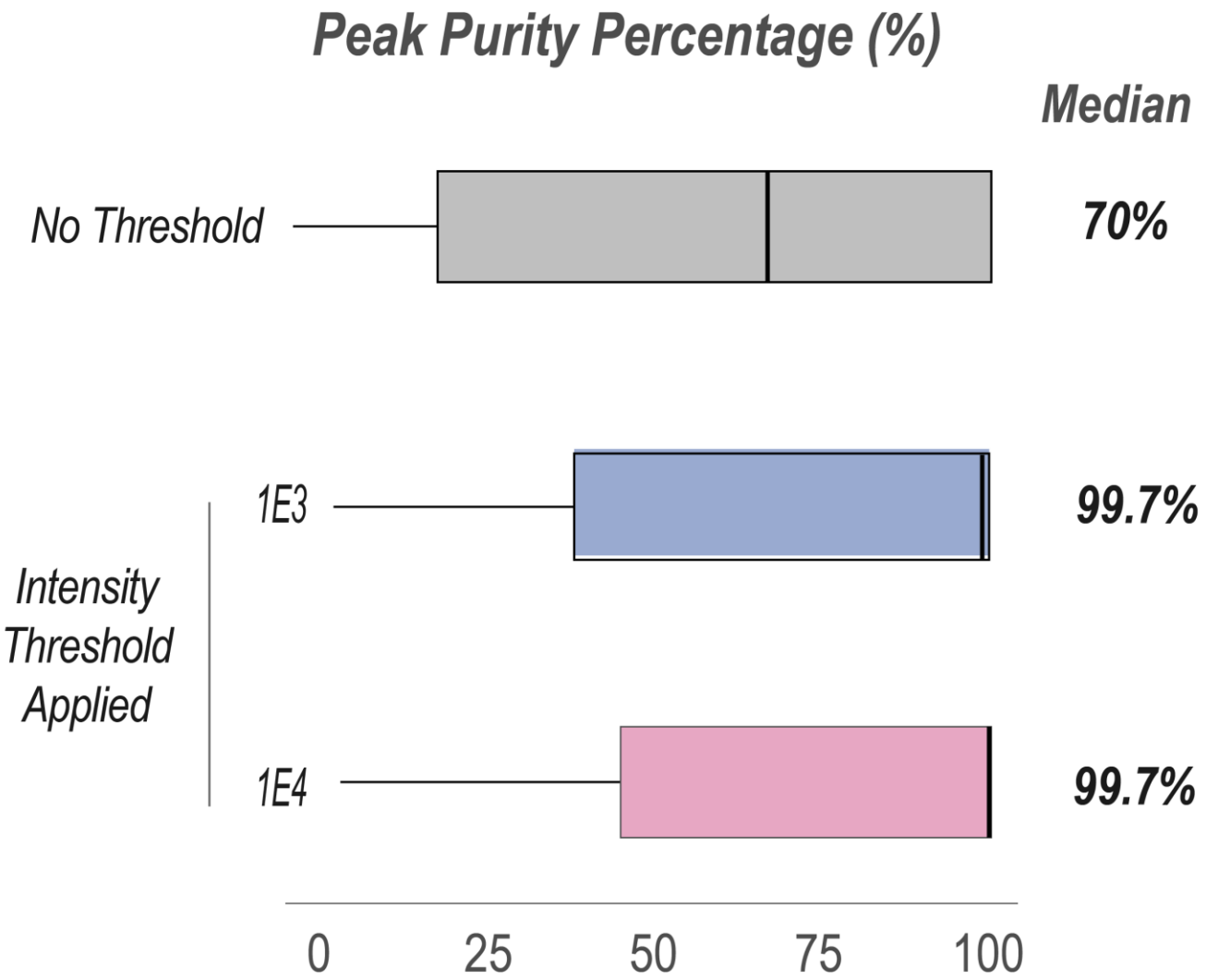
Intensity Threshold with Optimized Dynamic Exclusion Doubles Lipid IDs



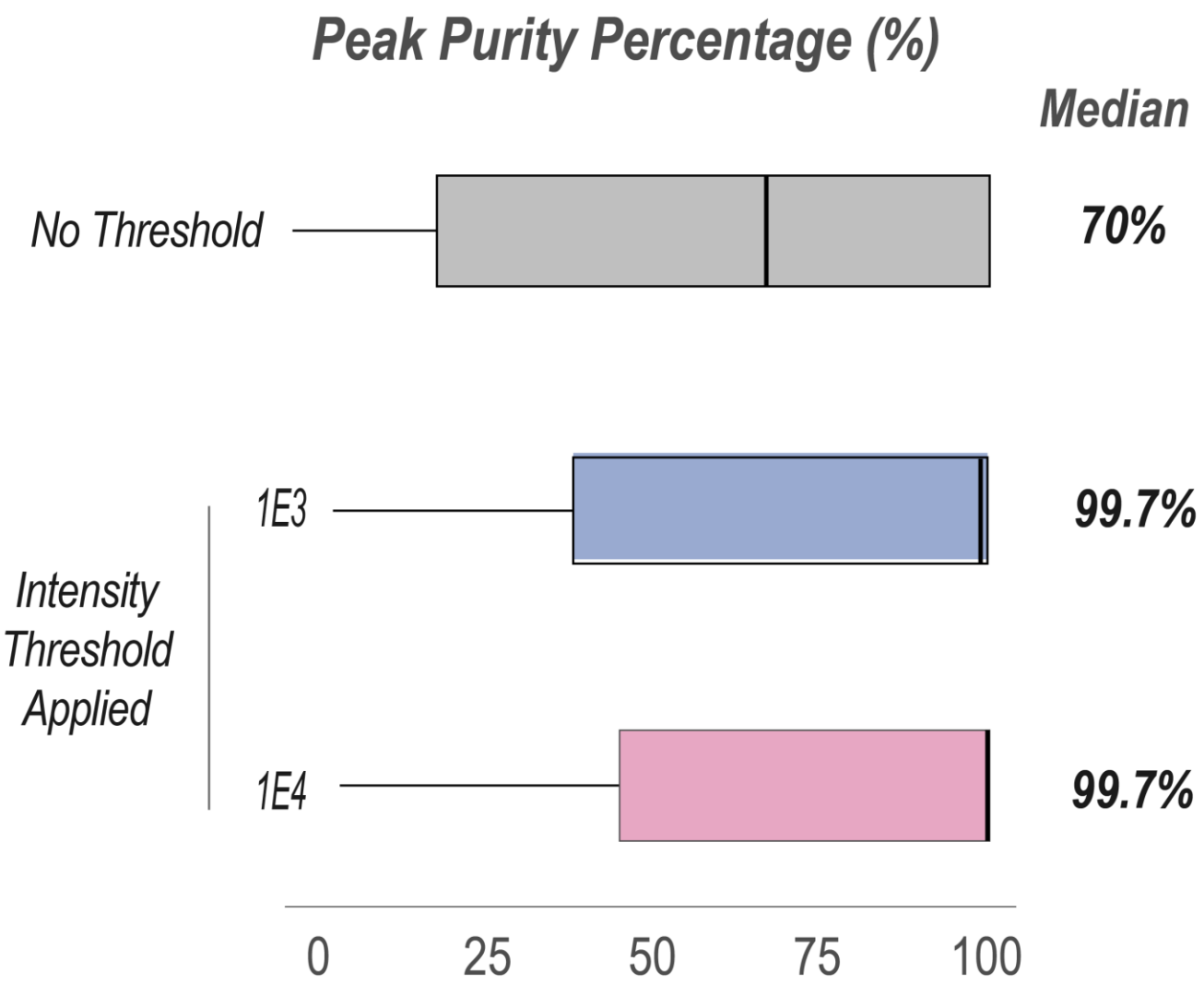
Intensity Threshold with Optimized Dynamic Exclusion Doubles Lipid IDs

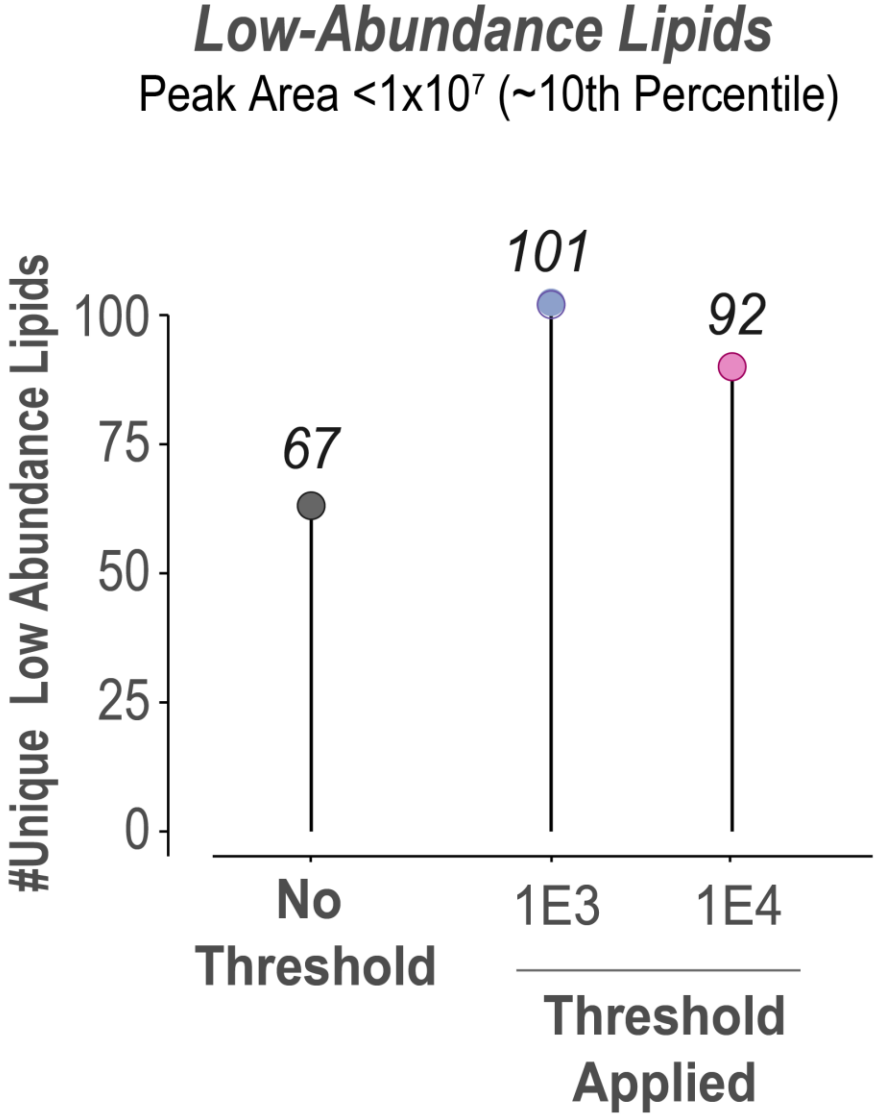


Intensity Threshold Improves Both Spectral Quality and Depth of Detection



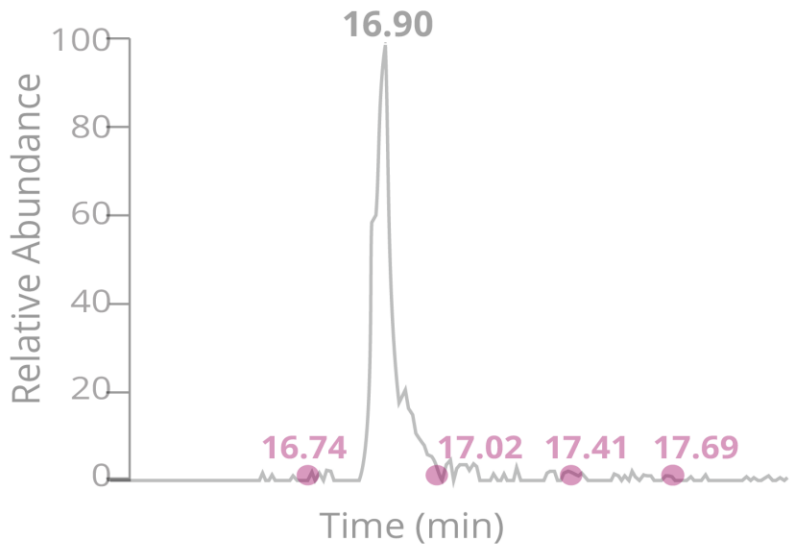
Intensity Threshold Improves Both Spectral Quality and Depth of Detection



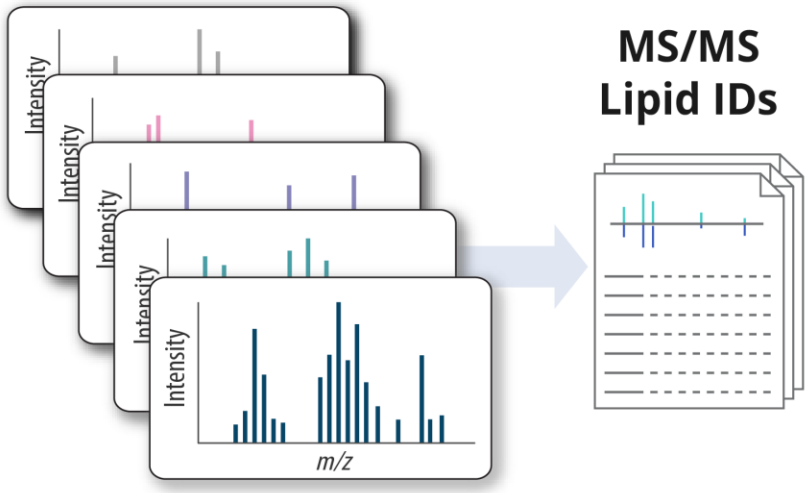


Low Number of Lipid Identifications

Sampling issues



Turning Scans into IDs

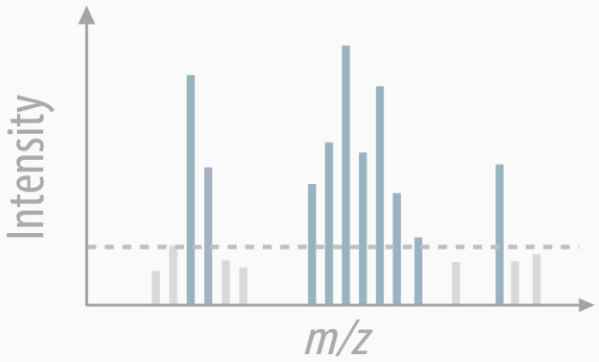


MS/MS Parameters to Optimize

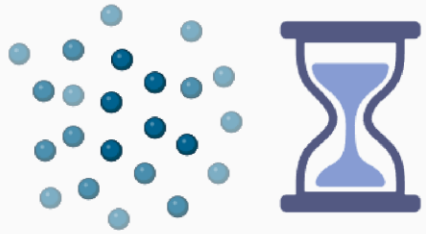
Dynamic Exclusion



Intensity Threshold

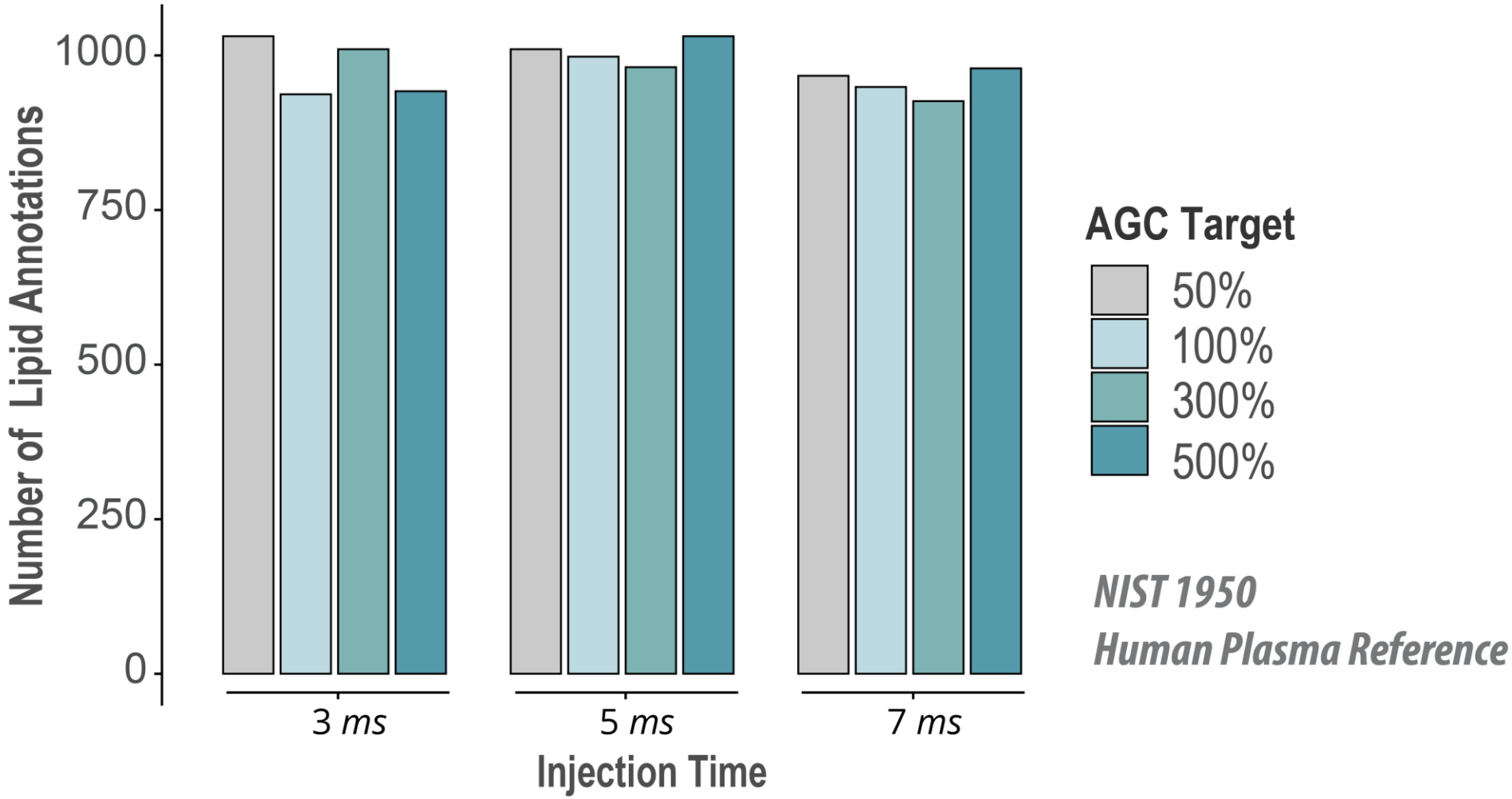


Automatic Gain Control and Max Injection Time



AGC and Injection Time Showed No Significant Change on Lipid Identifications

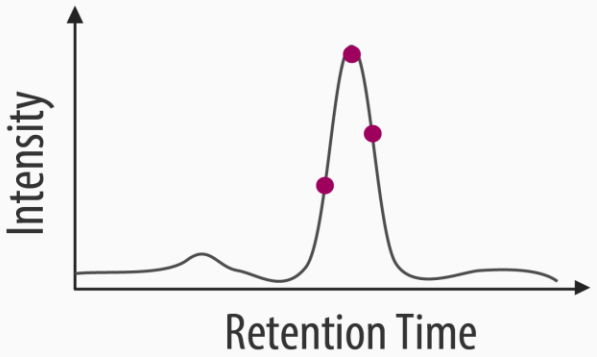
Automatic Gain Control (AGC) and Max Injection Time (MaxIT)



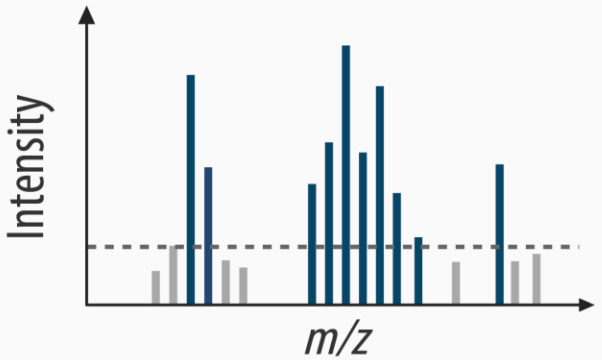
Summary

MS/MS Parameters Optimized

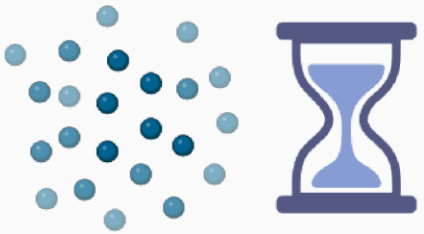
Dynamic Exclusion



Intensity Threshold

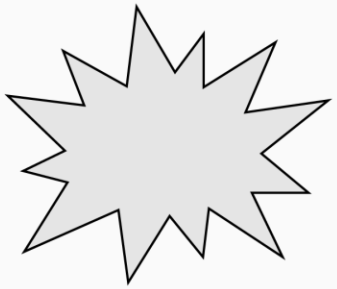


Automatic Gain Control and Max Injection Time

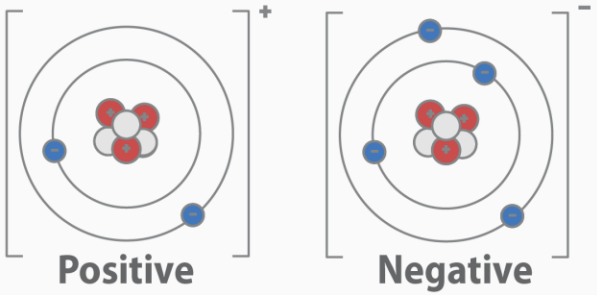


Future Directions

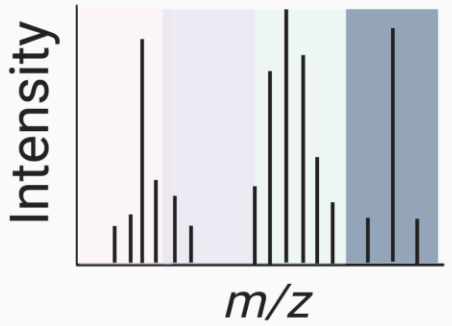
Stepped HCD Collision Energy



Ionization Modes



DIA



Investigating Caloric Restriction in a 30-Year Primate Aging Study

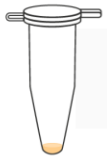
Control-Fed



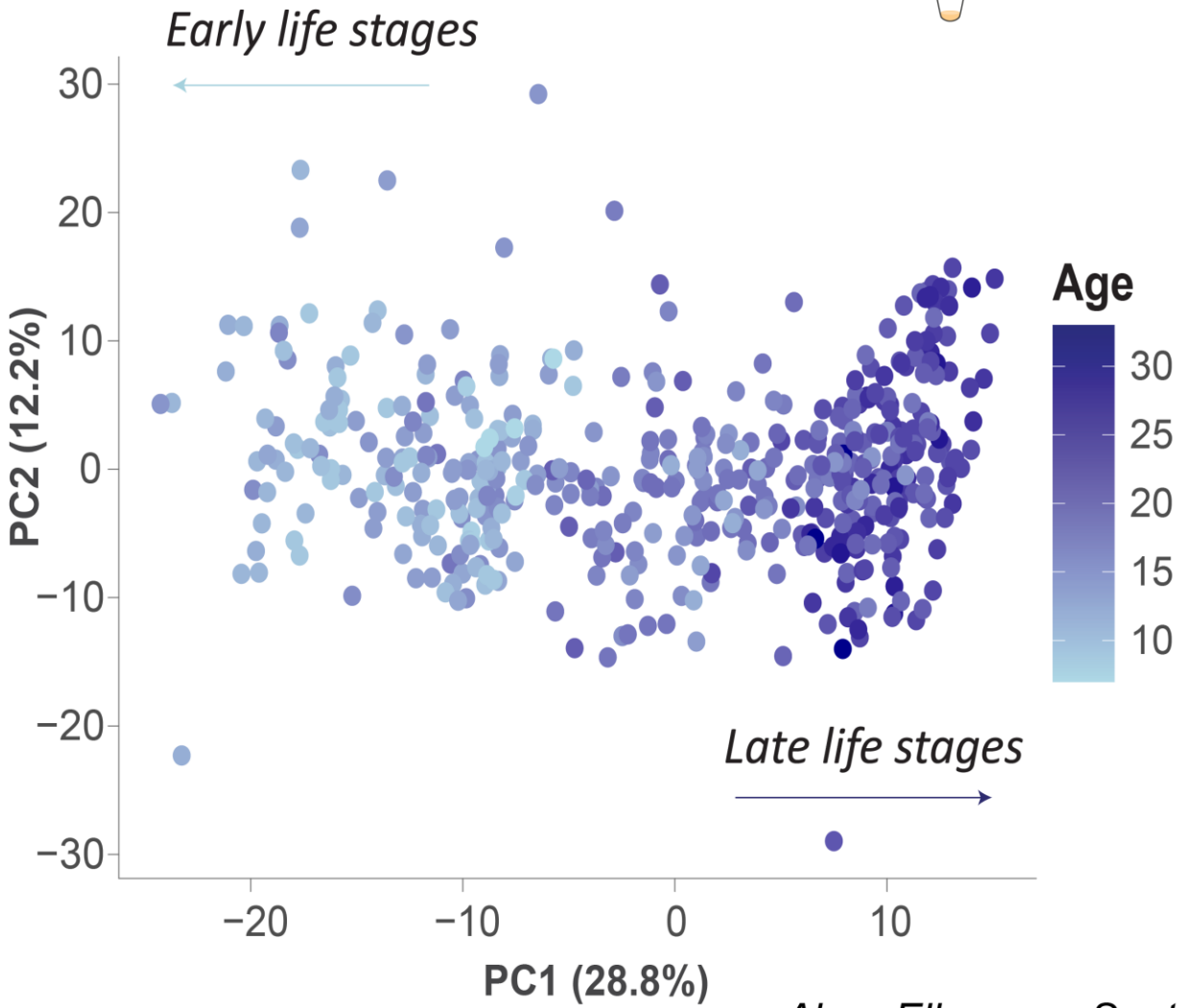
Caloric Restricted

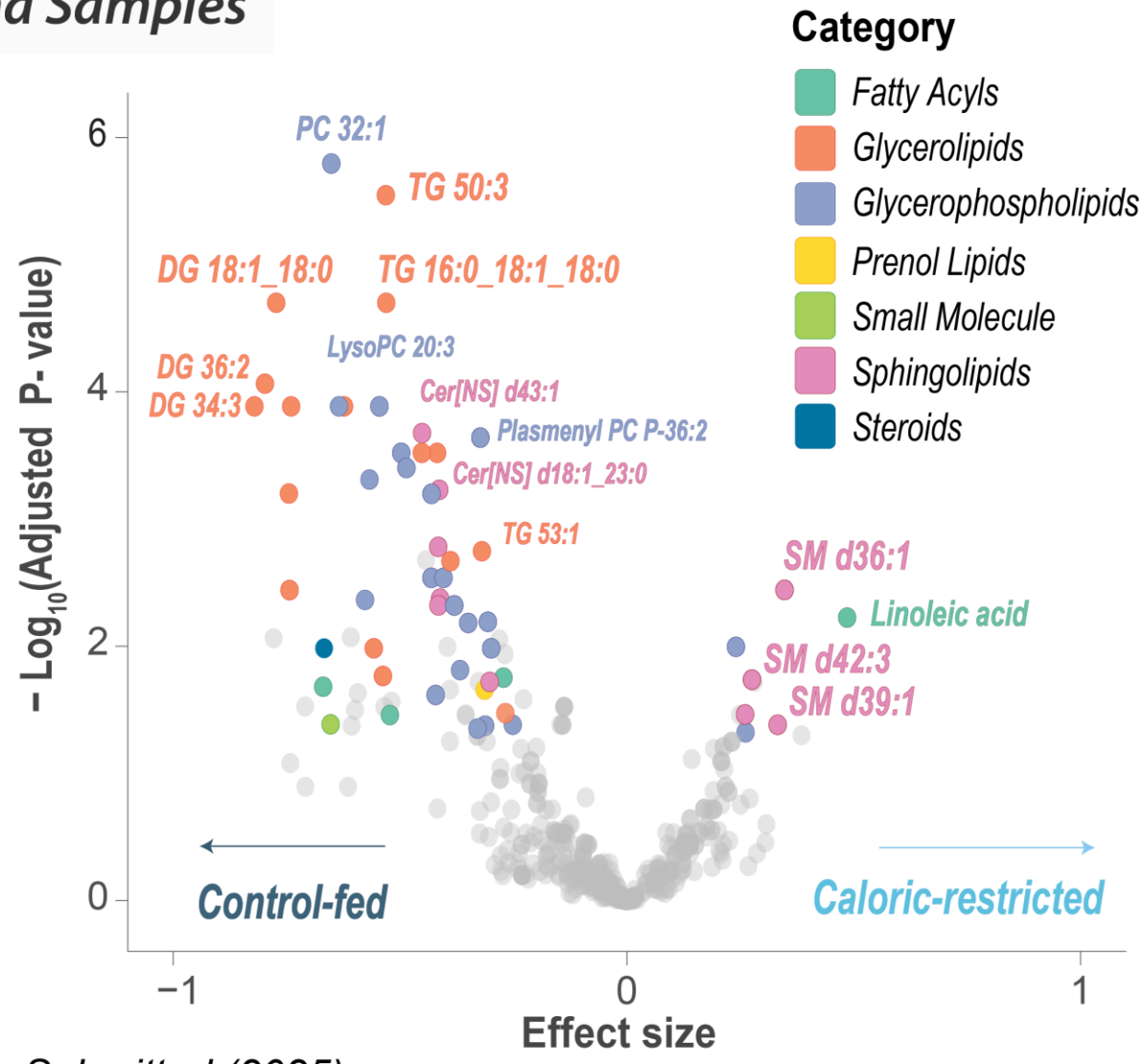
Colman, R. et. al., Science (2009)

Lipidomic Signatures Linked to Caloric Restriction and Aging



494 Plasma Samples

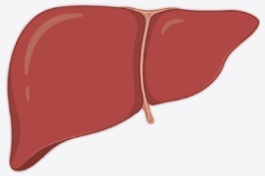




Abou Elhassan, S. et. al., Submitted (2025)

Exploring Caloric Restriction Impacts on Aging in Tissue Necropsies

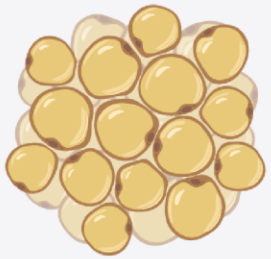
Liver



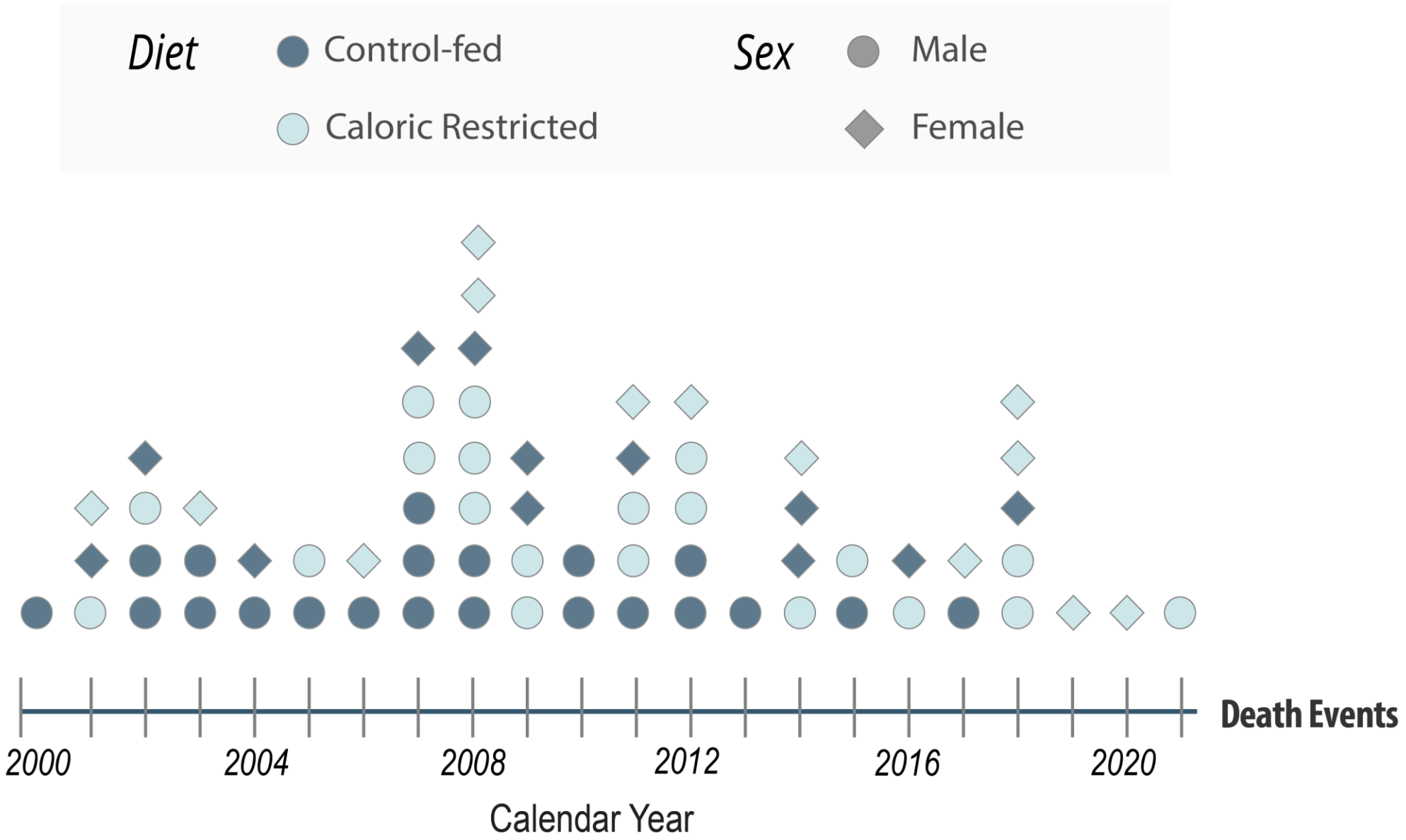
Muscle



Adipose



Tissue Necropsies




COON
LABORATORIES
NIH GM R35 118110 | NIH GM P41 108538
ThermoFisher
SCIENTIFIC
MORGRIDGE
INSTITUTE FOR RESEARCH
WISCONSIN
UNIVERSITY OF WISCONSIN–MADISON
LABORATORY FOR
BIOMOLECULAR
MASS SPECTROMETRY
NATIONAL CENTER FOR
QUANTITATIVE BIOLOGY
OF COMPLEX SYSTEMS